Exact Sidon Constants in Degrees Five, Six and Seven and Strict Monotonicity in Every Degree

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Abstract

For complex polynomials with consecutive frequencies, we identify S_5, S_6, S_7 by primitive integer minimal polynomials of degrees $4,372,855$ and explicit rational intervals containing exactly one root. We also prove analytically that $S_{n+1} > S_n$ for every $n \geq 1$. The individual exact closures combine complementary-polynomial feasibility, locally sharp Toeplitz bounds, complete finite phase covers, rational charts and exact rational-univariate identities. We give the analytic implications and verifier pseudocode in the paper; independent programs recheck all finite predicates. The degree-seven minimal polynomial has splitting-field Galois group $\mathfrak{S}_{855}$.

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1 Introduction and main results

For the consecutive spectrum $E_n = \{0, \dots, n\}$, define its Fourier-analytic Sidon constant by

$$S_n = \max\left\{ \mathcal{J}(P) : P(z) = \sum_{k=0}^{n} a_k z^k, \quad \|P\|_{\mathbb{T}} \le 1 \right\}, \qquad \mathcal{J}(P) = \sum_{k=0}^{n} |a_k|. \tag{1}$$

Here $\|P\|_{\mathbb{T}} = \max_{|z|=1} |P(z)|$; we also write $\|P\|_\infty$ for this norm. The coefficients are unrestricted complex numbers. Parseval bounds the coefficient vectors of the feasible set, which is closed, so the maximum exists. Degree means "at most" throughout this paper.

Small consecutive spectra already present a difficult exact optimization problem. Neuwirth's lecture manuscript discusses the classical Newman–Shapiro bound $S_n \le \sqrt{n}$ for $n \ge 1$, including equality in degrees $1, 2, 4$, and the degree-three problem [6]. Yu, Deepseek and Claude Code subsequently reported $S_3 = 5/3$ and a Lean formalization [9]. We cite that preprint for its stated result; our proofs and certificates do not depend on its formalization.

The present work determines three further constants as individual algebraic numbers. Here and below, the integer minimal polynomial means the unique primitive polynomial in $\mathbb{Z}[X]$ with positive leading coefficient associated with the monic minimal polynomial over $\mathbb{Q}$.

Theorem 1.1 (Exact closure of S_5)**.** *The integer minimal polynomial of S_5 is*

$$h_5(X) = 675X^4 - 1026X^3 - 936X^2 - 414X + 677. \tag{2}$$

It has exactly one root in $I_5 = (218/100, 219/100)$, and that root is S_5. In particular, $[\mathbb{Q}(S_5) : \mathbb{Q}] = 4$ and $S_5 = 2.186529771892569\ldots$. A real extremizer is given explicitly in Section 4; the upper bound holds for all complex coefficients.

Theorem 1.2 (Exact closure of S_6)**.** *The constant S_6 is the unique root in*

$$I_6 = \left(\frac{2362563342647160829}{10^{18}}, \frac{2362563342647160830}{10^{18}}\right) \tag{3}$$

of the primitive integer polynomial h_6 supplied in `S6_S7_Minimal_Polynomials.txt`. This polynomial is irreducible, has degree 372, and equals $H_6(X^2)$ with $\deg H_6 = 186$. Thus $[\mathbb{Q}(S_6) : \mathbb{Q}] = 372$ and $[\mathbb{Q}(S_6^2) : \mathbb{Q}] = 186$.

Theorem 1.3 (Exact closure of S_7)**.** *The constant S_7 is the unique root in*

$$I_7 = \left(\frac{2551658061766431151}{10^{18}}, \frac{2551658061766431152}{10^{18}}\right) \tag{4}$$

of the primitive integer polynomial h_7 supplied in the same text file. The polynomial is irreducible of degree 855. Consequently $[\mathbb{Q}(S_7) : \mathbb{Q}] = [\mathbb{Q}(S_7^2) : \mathbb{Q}] = 855$.

Theorem 1.4 (Strict growth)**.** *One has $S_0 = S_1 = 1$, and*

$$S_{n+1} > S_n \qquad (n \geq 1). \tag{5}$$

This assertion has a purely analytic proof, independent of the finite certificates for the three individual constants.

The decimal enclosures in (3)–(4) are rational open intervals of width 10^{-18}. The long coefficient lists of h_6, h_7 are the only polynomial displays deferred to the separate text file. Its unambiguous coefficient convention is $h_n(X) = \sum_{j=0}^{\deg h_n} c_{n,j} X^j$, with all integer coefficients listed in increasing order, including zeros. The certificate compares these lists with the defining rational-univariate data coefficient by coefficient.

Proof architecture and relation to previous methods. The argument has three independent obligations. First, a complementary polynomial proves feasibility and positive Toeplitz moments give a sharp *global* upper bound after a complete phase cover. Second, a rational chart and a contracting six-variable box identify that unrestricted global extremizer with a specified real branch of a rational univariate representation (RUR). Third, integer identities, rational root isolation and irreducibility turn an annihilating polynomial into a minimal polynomial. None of the latter two obligations is discharged by a small floating-point residual or by a multivariate isolation box alone.

The analytic tools have established antecedents. Scalar Fejér–Riesz factorization and its operator extensions are surveyed in [2]; the matrix-product charts below are a symmetric paraunitary specialization related to quantum signal processing [3]. RUR methods originate well before this work [8]. We prove the particular inverse formulas and root-identification statements needed here. The local Taylor–Schur certificates complement global trigonometric optimization approaches such as [1]; that work is background, not an external optimality certificate for our candidates. Irreducibility is verified with finite-field Frobenius tests of the type introduced by Rabin [7].

The proof programs have two arithmetic layers. The complete phase covers and the degree-five field computations use Python's standard-library integer arithmetic. The new univariate and RUR checks use exact FLINT integer polynomials and outward-rounded Arb balls through `python-flint`; see [4]. SymPy constructs exact rational expressions. Neither an optimizer, a stored success flag, nor an external system solver is trusted by the verifier. Appendix D gives the full execution interface and mathematical pseudocode.

2 Complementary factors and strict monotonicity

We record the scalar factorization in the form used in both the analytic and computer-assisted parts of the paper.

Lemma 2.1 (Scalar complementary factor)**.** *If P has degree at most n and $\|P\|_{\mathbb{T}} \le 1$, there is a polynomial Q of degree at most n with $|P|^2 + |Q|^2 = 1$ on $\mathbb{T}$. If $|P|$ is not identically one, Q can be chosen to have degree exactly n and nonzero leading coefficient.*

Proof. Put $D(z) = 1 - P(z)\overline{P(1/\bar{z})}$. If it vanishes identically take $Q = 0$. Otherwise let d be its actual Laurent degree. For $d = 0$ the assertion is immediate. For $d > 0$, $z^d D(z)$ has degree $2d$ with nonzero constant and leading terms. Its zeros off $\mathbb{T}$ occur in pairs $\alpha, 1/\bar{\alpha}$ with equal multiplicities. A zero on $\mathbb{T}$ has even multiplicity: the real-analytic function $D(e^{it}) \ge 0$ cannot change sign at its zero. Select one zero from each off-circle pair and half the multiplicity of every circle zero to form a degree-d polynomial q. Comparing zeros gives $D = c|q|^2$ on $\mathbb{T}$, with $c > 0$ by evaluation at a point where $D > 0$. Absorb $\sqrt{c}$ in q. Finally, $Q = z^{n-d}q$ leaves its modulus unchanged and has degree n. □

Lemma 2.2 (A degree-raising perturbation)**.** *Let P be a nonmonomial polynomial of degree at most n with $\|P\|_{\mathbb{T}} = 1$. There exists a polynomial $\widetilde{P}$ of degree at most $n+1$, with $\|\widetilde{P}\|_{\mathbb{T}} \le 1$ and $\mathcal{J}(\widetilde{P}) > \mathcal{J}(P)$.*

Proof. A polynomial of constant modulus one on $\mathbb{T}$ is a monomial. Indeed, if its lowest and highest nonzero coefficients have indices $\ell < r$, the autocorrelation at lag $r - \ell$ equals $a_r \bar{a}_\ell \neq 0$, contradicting constant modulus. Thus Lemma 2.1 gives a nonzero Q of degree n, with leading coefficient $q_n \neq 0$ and $|P|^2 + |Q|^2 = 1$.

For $0 < \varepsilon < 1$ and $|\eta| = 1$, set

$$v = (\varepsilon, -\sqrt{1-\varepsilon^2}\,\bar{\eta})^T, \quad \Pi = vv^*, \quad E_\Pi(z) = I - \Pi + z\Pi.$$

On $\mathbb{T}$, E_Π is unitary, since Π is an orthogonal projection. Its action on $(P, Q)^T$ therefore preserves the pointwise squared norm. The first component is the feasible polynomial

$$P_\varepsilon = (1 - \varepsilon^2 + \varepsilon^2 z)P + \varepsilon\sqrt{1-\varepsilon^2}\,\eta(1-z)Q. \tag{6}$$

Write $R = (1-z)Q$, and append the coefficient $a_{n+1} = 0$ to P. In the finite-dimensional coefficient space, $P_\varepsilon = P + \varepsilon\eta R + O(\varepsilon^2)$, whence

$$\left.\frac{d}{d\varepsilon}\right|_{0+} \mathcal{J}(P_\varepsilon) = \mathrm{Re}\left(\eta \sum_{a_k \neq 0} R_k \frac{\bar{a}_k}{|a_k|}\right) + \sum_{a_k = 0} |R_k|. \tag{7}$$

This follows term by term from the directional derivative of the absolute value, including $|0 + \varepsilon\eta R_k|$ at a zero coefficient. Choose η so that the first real part is zero (any η works if its sum is zero). The second sum is at least $|R_{n+1}| = |q_n| > 0$. Hence $\mathcal{J}(P_\varepsilon) > \mathcal{J}(P)$ for sufficiently small positive ε. □

Proof of Theorem 1.4. For degree one, the triangle inequality is attained by a suitable point of $\mathbb{T}$, so $\|a_0 + a_1 z\|_{\mathbb{T}} = |a_0| + |a_1|$; this also gives $S_0 = S_1 = 1$. Every maximizer has norm one: $S_n \geq 1$, and a nonzero maximizer of smaller norm could be rescaled to increase $\mathcal{J}$. For $n \geq 1$ a nonmonomial maximizer can always be chosen. If $S_n > 1$, every maximizer is nonmonomial. If $S_n = 1$, the polynomial $(1 + z)/2$ is a nonmonomial maximizer. Apply Lemma 2.2 to obtain $S_{n+1} > S_n$. □

Proposition 2.3 (Algebraicity for every fixed degree)**.** *Each S_n is real algebraic, and has an extremizer whose real and imaginary coefficient parts are real algebraic.*

Proof. Use Lemma 2.1 and equate coefficients in $|P|^2 + |Q|^2 = 1$:

$$\sum_{k=0}^{n-h} (a_{k+h}\bar{a}_k + q_{k+h}\bar{q}_k) = \mathbf{1}_{h=0} \quad (0 \leq h \leq n).$$

Add $\rho_k \geq 0$, $\rho_k^2 = |a_k|^2$. In real and imaginary parts this is a compact semialgebraic set over $\mathbb{Q}$; the zero-lag equation bounds every coordinate. Its image under $\sum \rho_k$ is a compact semialgebraic subset of $\mathbb{R}$ over $\mathbb{Q}$, whose rightmost endpoint is algebraic. Real-closed-field transfer supplies a real-algebraic point of the nonempty extremal fiber. This is an existence statement, not an efficient elimination algorithm. □

3 Phase functionals and positive Toeplitz duality

For a fixed degree n, and $b = (b_0, \ldots, b_n) \in \mathbb{T}^{n+1}$ and $P(z) = \sum a_k z^k$, define

$$\Phi_b(P) = b^T a = \sum_{k=0}^{n} b_k a_k, \qquad L(b) = \max_{\|P\|_\infty \leq 1} |\Phi_b(P)|, \qquad R_b = \bar{b}\, b^T. \tag{8}$$

Here $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, transpose T does not mean conjugate transpose, and R_b is the Hermitian matrix satisfying $a^* R_b a = |b^T a|^2$. Since $\sum |a_k| = \max_{b \in \mathbb{T}^{n+1}} |b^T a|$,

$$S_n = \max_{b \in \mathbb{T}^{n+1}} L(b). \tag{9}$$

Lemma 3.1 (Finite positive Toeplitz moments)**.** *Suppose $T = (t_{k-j})_{j,k=0}^n$ is Hermitian Toeplitz and positive semidefinite. There exist finitely many $\zeta_l \in \mathbb{T}$ and $\mu_l \geq 0$ such that*

$$T_{jk} = \sum_l \mu_l \overline{\zeta_l}^{\,j} \zeta_l^k, \qquad \sum_l \mu_l = t_0. \tag{10}$$

Consequently, $a^* T a \leq t_0 \|P\|_\infty^2$ for the coefficient vector a of any polynomial of degree at most n.

Proof. For $n = 0$, take the single atom $\zeta_1 = 1$ of weight t_0. For $n \geq 1$, choose Gram vectors $v_0, \ldots, v_n$ satisfying $\langle v_j, v_k \rangle = T_{jk}$, with the inner product linear in its second argument. Toeplitzness implies that $v_j \mapsto v_{j+1}$ is a well-defined isometry from $\operatorname{span}(v_0, \ldots, v_{n-1})$ onto $\operatorname{span}(v_1, \ldots, v_n)$: any linear relation is preserved, because the norms of corresponding linear combinations agree. The two subspaces have the same finite dimension. Extending this isometry by an arbitrary isometry between their orthogonal complements produces a unitary operator U on $\operatorname{span}(v_0, \ldots, v_n)$. Then $v_j = U^j v_0$. The spectral decomposition of the finite-dimensional unitary U gives spectral values $\zeta_l \in \mathbb{T}$ and orthogonal spectral projections E_l. Taking $\mu_l = \|E_l v_0\|^2$ proves (10). In particular, $a^* T a = \sum_l \mu_l |P(\zeta_l)|^2 \leq t_0 \|P\|_\infty^2$. □

Proposition 3.2 (Toeplitz domination)**.** *If T is Hermitian Toeplitz and $T - R_b \succeq 0$, then*

$$L(b)^2 \le T_{00}. \tag{11}$$

Moreover, $L(b)^2$ equals the minimum of T_{00} over all such matrices T.

Proof. Since $R_b \succeq 0$, the hypothesis implies $T \succeq 0$. By Lemma 3.1, $|\Phi_b(P)|^2 = a^* R_b a \le a^* T a \le T_{00}\|P\|_\infty^2$, proving (11). For the reverse equality, extend the coefficient functional Φ_b from the $(n+1)$-dimensional polynomial subspace of $C(\mathbb{T})$ to $C(\mathbb{T})$ without changing its norm, by complex Hahn–Banach. The Riesz representation theorem supplies a complex measure ν of total variation $L(b)$ representing this extension. Set $T_{jk} = L(b)\int_{\mathbb{T}} \overline{z}^j z^k \, d|\nu|(z)$. This matrix is Hermitian Toeplitz, with $T_{00} = L(b)^2$, and Cauchy–Schwarz for $|\nu|$ gives $|\Phi_b(P)|^2 \le L(b)\int |P|^2 d|\nu| = a^* T a$. Thus $T - R_b \succeq 0$. (The case $L(b) = 0$ is immediate.) □

3.1 The exact real-amplitude dual

For $n \ge 1$, put $m = n+1$, and for $b \in \mathbb{T}^m$ let

$$\Phi_b(P) = \sum_{k=0}^{n} b_k a_k, \quad L_n(b) = \sup_{\|P\|_{\mathbb{T}} \le 1} |\Phi_b(P)|, \quad M_n(b) = \sup_{\substack{x \in \mathbb{R}^m \\ \|\sum_{k=0}^n \bar{b}_k x_k z^k\|_{\mathbb{T}} \le 1}} \left| \sum_{k=0}^{n} x_k \right|.$$

Clearly $M_n(b) \le S_n$ and, by choosing $b_k = \bar{a}_k/|a_k|$ and $x_k = |a_k|$ (with an arbitrary phase for a zero coefficient), $\max_b M_n(b) = S_n$. In particular, a sharp bound for M_n in a neighborhood of a candidate phase is sufficient to control all *aligned* extremizers in that neighborhood.

Theorem 3.3 (Exact real-amplitude Toeplitz duality)**.** *With $D_b = \operatorname{diag}(b_0, \dots, b_n)$ and $\mathbf{1} = (1, \dots, 1)^T$, one has*

$$M_n(b)^2 = \min\left\{ T_{00} : \; T = T^* \succeq 0 \text{ is Toeplitz}, \quad \operatorname{Re}(D_b T D_{\bar{b}}) - \mathbf{1}\mathbf{1}^T \succeq 0 \text{ on } \mathbb{R}^m \right\}. \tag{12}$$

Proof. The finite positive Toeplitz moment representation established in Section 3 writes $T_{jk} = \int_{\mathbb{T}} \overline{z}^j z^k \, d\nu(z)$ with $\nu \ge 0$ and $\nu(\mathbb{T}) = T_{00}$. The matrix inequality in (12), tested on a real vector x, yields

$$|\mathbf{1}^T x|^2 \le x^T \operatorname{Re}(D_b T D_{\bar{b}}) x = \int_{\mathbb{T}} \left| \sum \bar{b}_k x_k z^k \right|^2 d\nu \le T_{00} \left\| \sum \bar{b}_k x_k z^k \right\|_{\mathbb{T}}^2.$$

For the opposite inequality, apply real Hahn–Banach to the real-linear functional $\sum x_k$ on the real subspace of polynomials $\sum \bar{b}_k x_k z^k$. Extend it without changing its norm $M = M_n(b)$ to $C(\mathbb{T}; \mathbb{C})$ considered as a real Banach space. The real Riesz representation has the form $P \mapsto \operatorname{Re}\int P \, d\mu$ for a complex measure of total variation M. Cauchy–Schwarz shows

$$|\mathbf{1}^T x|^2 \le \left| \int P \, d\mu \right|^2 \le M \int |P|^2 \, d|\mu|.$$

The Toeplitz matrix of $M|\mu|$ has diagonal M^2, is positive, and obeys the required real matrix inequality. When $M = 0$, use $T = 0$. □

The condition $T \succeq 0$ must be retained: positivity of only the real matrix difference does not imply that T is a positive Toeplitz moment matrix. The certificates below avoid this trap by constructing T directly from *positive moving atoms*. The complex majorant $T - \bar{b}b^T \succeq 0$ continues to be useful away from the sharp candidate, where strictness can be purchased cheaply.

4 Degree five: an explicit quartic value and a global proof

Let w be the unique real root in $(1/8, 13/100)$ of

$$f(w) = 16w^4 + 10w^3 - 8w + 1 = 0. \tag{13}$$

Define the six real numbers $p_0, \dots, p_5$ by

$$p_k = \frac{A_k + B_k w + C_k w^2 + D_k w^3}{25875}, \tag{14}$$

where

k	A_k	B_k	C_k	D_k
0	3112	4752	9708	-9248
1	9870	-6480	-28920	-12480
2	13460	-4590	-34860	-31840
3	7780	11880	24270	-23120
4	-15705	-2430	32280	64320
5	7358	-3132	-2478	12368

Set $P_*(z) = \sum_{k=0}^{5} p_k z^k$ and

$$\Lambda = \frac{3819 + 324w - 4304w^2 - 8576w^3}{1725}. \tag{15}$$

Proposition 4.1 (An explicit extremizer and its full orbit)**.** *For every polynomial $P(z) = \sum_{k=0}^{5} a_k z^k$ with complex coefficients,*

$$\sum_{k=0}^{5} |a_k| \le \Lambda \max_{|z|=1} |P(z)|, \tag{16}$$

where

$$\Lambda = 2.186529771892569\ldots$$

is the unique root in $(2.18, 2.19)$ of h_5 in (2). Equality for a nonzero polynomial holds if and only if, after normalizing its supremum norm to one, it has one of the forms

$$\gamma P_*(\omega z), \qquad \gamma z^5 P_*(\omega/z), \qquad |\gamma| = |\omega| = 1. \tag{17}$$

Consequently $S_5 = S_5^{\mathbb{R}} = \Lambda$, where $S_5^{\mathbb{R}}$ denotes the analogous constant restricted to real coefficients.

4.1 An exact algebraic extremizer

The polynomial (13) has opposite signs at $1/8$ and $13/100$. Its derivative $64w^3 + 30w^2 - 8$ is negative throughout this interval, hence w is uniquely defined. The companion certificate uses rational bisection to enclose w and checks that $p_0, p_1, p_2, p_3, p_5 > 0$ and $p_4 < 0$. Thus with

$$s = (1, 1, 1, 1, -1, 1)^T, \tag{18}$$

we have $s^T p = \sum |p_k| = \Lambda$. The program also verifies $P_*(1) = 1$ and the quartic identity (2), by reduction modulo $f(w)$ in $\mathbb{Q}[w]$; uniqueness of the root in $(2.18, 2.19)$ follows because the polynomial in (2) is strictly increasing there and changes sign at the endpoints.

Here is a direct unit-circle feasibility identity. Put

$$\eta = \frac{p_3 - p_0}{p_5}, \qquad \kappa = 1 - \eta, \qquad Q_*(z) = \sqrt{p_0 p_5}\,(1 - \kappa z^2 + \kappa z^3 - z^5). \tag{19}$$

The expressions are well-defined since $p_0 p_5 > 0$. One has, for $h = 0, \dots, 5$,

$$\sum_{k=0}^{5-h} \bigl(p_k p_{k+h} + q_k q_{k+h}\bigr) = \begin{cases} 1, & h = 0, \\ 0, & h > 0, \end{cases} \tag{20}$$

where $Q_*(z) = \sum_{k=0}^{5} q_k z^k$. Indeed, $q_k q_j = p_0 p_5 v_k v_j$ with $v = (1, 0, -\kappa, \kappa, 0, -1)^T$; each of the six identities is an identity of rational functions of w, and the rational reductions in the certificate check them exactly, without any square-root approximations. Equating Fourier coefficients in (20) gives

$$|P_*(z)|^2 + |Q_*(z)|^2 = 1 \qquad (|z| = 1). \tag{21}$$

Therefore $\|P_*\|_\infty \le 1$ and $\mathcal{J}(P_*) = \Lambda$. Since $P_*(1) = 1$, its norm is exactly one. We have proved

$$S_5 \ge \Lambda. \tag{22}$$

This explicit complementary-square factorization performs two tasks: it verifies primal feasibility and, through its autocorrelation identities, supplies the trace relation needed in the sharp local dual construction.

4.2 Normalized degree-five phases

We abbreviate $L_5(b)$ by $L(b)$ in this section.

The symmetries

$$b_k \mapsto \gamma \omega^k b_k, \quad b \mapsto \overline{b}, \quad (b_0, \dots, b_5) \mapsto (b_5, \dots, b_0), \qquad |\gamma| = |\omega| = 1, \tag{23}$$

preserve $L(b)$, by scalar multiplication, rotation of the variable, conjugation, and reversal of the polynomial. Choosing γ and a fifth root ω allows $b_0 = b_5 = 1$. We therefore parameterize all normalized phase classes by

$$b(\theta) = (1, e^{i\theta_1}, e^{i\theta_2}, e^{i\theta_3}, e^{i\theta_4}, 1), \qquad \theta \in (\mathbb{R}/2\pi\mathbb{Z})^4. \tag{24}$$

The normalized phases associated with the orbit of P_* and its reversal are precisely the ten points

$$(0,0,0,\pi) + \frac{2\pi r}{5}(1,2,3,4), \quad (\pi,0,0,0) + \frac{2\pi r}{5}(1,2,3,4), \qquad 0 \le r \le 4, \tag{25}$$

with angles taken modulo 2π.

4.3 A locally sharp continuous Toeplitz certificate

This section proves an exact, not approximate, phase bound near each point in (25). Several small algebraic inequalities are discharged by the accompanying certificate, using fully specified matrices. We distinguish these inputs from the analytic perturbation argument.

4.3.1 The sharp base point and its nullspace

Let $p = (p_k)_{k=0}^5$, $q = (q_k)_{k=0}^5$, and s be as in (14), (19), and (18). Write

$$N = [p\ q] \in \mathbb{R}^{6\times 2}, \qquad \Pi = N(N^TN)^{-1}N^T. \tag{26}$$

Thus Π is the orthogonal projection onto $\operatorname{span}\{p, q\}$, provided N^TN is invertible, as certified below. Define $t_0 = \Lambda^2$ and t_1, t_2 by

$$At_1 + Bt_2 = \Lambda - t_0\{p_0 - \kappa p_3 + (1-\kappa^2)p_5\}, \tag{27}$$
$$Ct_1 + Dt_2 = \Lambda - t_0\{(1-\kappa^2)p_0 - \kappa p_2 + p_5\}, \tag{28}$$

where

$$A = p_1 + \kappa p_3 + \eta p_4 + \kappa^2 p_5, \qquad B = p_2 + p_3 + \kappa p_4,$$
$$C = \kappa^2 p_0 + \eta p_1 + \kappa p_2 + p_4, \qquad D = \kappa p_1 + p_2 + p_3.$$

Finally set

$$t_3 = t_2 - \kappa t_0 + \kappa t_1, \qquad t_4 = \eta t_1 + \kappa t_2, \qquad t_5 = (1-\kappa^2)t_0 + \kappa^2 t_1, \tag{29}$$

and let $T_* = (t_{|j-k|})_{j,k=0}^5$. The determinant of the system (27)–(28) is nonzero; it is certified by rational interval arithmetic in $\mathbb{Q}(w)$.

Lemma 4.2 (Base-point algebraic certificate)**.** *With $S_0 = T_* - ss^T$, one has*

$$T_*p = \Lambda s, \qquad T_*q = 0, \qquad S_0N = 0, \qquad S_0 + 5\Pi - 4I_6 \succ 0. \tag{30}$$

Consequently $S_0 \succeq 0$, $\ker S_0 = \operatorname{ran} N$, and its restriction to $(\operatorname{ran} N)^\perp$ is strictly greater than $4I$.

Proof. The first two identities reduce to exact identities in $\mathbb{Q}(w)$ after using (27)–(29); the quartic-field part of the certificate checks all twelve components. Moreover $s^Tp = \Lambda$ and $s^Tq = 0$, so these identities imply $S_0N = 0$. The certificate checks $N^TN \succ 0$ and the last matrix inequality in (30) by outward rational intervals and positive LDL^T pivots. Since S_0 is symmetric, it preserves the decomposition $\operatorname{ran} N \oplus (\operatorname{ran} N)^\perp$. On the first summand it is zero, while on the second $\Pi = 0$ and the last inequality gives $S_0 > 4I$. This proves all claims. □

In particular, $T_* - R_s \succeq 0$ is a sharp dual certificate for Φ_s, because $\Phi_s(P_*) = \Lambda$ and $(T_*)_{00} = \Lambda^2$. The matrix S_0 has an unavoidable two-dimensional nullspace. Put

$$G = (S_0 + \Pi)^{-1} - \Pi. \tag{31}$$

Then G is the inverse of S_0 on $(\operatorname{ran} N)^\perp$, extended by zero on $\operatorname{ran} N$, and $\|G\|_{\mathrm{op}} < 1/4$. By (20) at $h = 0$, $\|N\|_{\mathrm{op}} \le \|N\|_F = 1$.

4.3.2 First-order correction and a strict quadratic form

Let $\theta_0 = \theta_5 = 0$ and $\|\theta\|_2 \le 1/125$. Put $x_i = \sin\theta_i$ for $1 \le i \le 4$, $r = \|x\|_2$, and $\widehat{x} = (0, x_1, \dots, x_4, 0)^T$. We write $u \odot v$ for componentwise multiplication and define

$$v = s \odot \widehat{x}, \qquad w_2 = s \odot \widehat{x}^{\,2}, \qquad R_1 = sv^T - vs^T, \qquad R_2 = vv^T - \tfrac{1}{2}(sw_2^T + w_2s^T). \tag{32}$$

The power $\widehat{x}^{\,2}$ is componentwise. Let J_h $(1 \le h \le 5)$ be the real skew-symmetric Toeplitz matrix with $+1$ on the hth upper diagonal and -1 on the hth lower diagonal. Let E_h be the real symmetric Toeplitz matrix with $+1$ on both such diagonals and zero elsewhere.

For each x, choose the unique $A_1(x) = \sum_{h=1}^{5} \alpha_h(x) J_h$ minimizing

$$\operatorname{tr}\big(((A_1 - R_1)N)^T G((A_1 - R_1)N)\big) \tag{33}$$

subject to

$$p^T (A_1 - R_1) q = 0. \tag{34}$$

The five coefficients are linear in x. Indeed, the objective in (33) is a nonnegative quadratic form, and the bordered Karush–Kuhn–Tucker system below is nonsingular by a certified exact interval test. Its solution is the unique constrained minimizer: any other minimizer would give a nonzero vector in the kernel of this bordered system by subtracting the two stationary equations. For an explicit, reproducible definition, put

$$c_h = p^T J_h q, \qquad M_{h\ell} = \operatorname{tr}\big((J_h N)^T G(J_\ell N)\big), \qquad \mathcal{B} = \begin{pmatrix} M & c \\ c^T & 0 \end{pmatrix}. \tag{35}$$

For $v = s \odot \widehat{x}$, let $r_h(x) = \operatorname{tr}((J_h N)^T G(R_1 N))$ and $d(x) = p^T R_1 q = \Lambda v^T q$. Then $\binom{\alpha(x)}{\tau(x)} = \mathcal{B}^{-1}\binom{r(x)}{d(x)}$ are precisely the constrained normal equations; $\mathcal{B}$ is verified invertible by the certificate. Define $K(x) = A_1(x) - R_1$, and write

$$K(x) = \sum_{i=1}^{4} x_i K_i. \tag{36}$$

As K is skew-symmetric, $N^T K N = 0$ follows from (34).

Define the real symmetric 4×4 matrix H by its quadratic form

$$x^T H x = \Lambda \sum_{i=1}^{4} |p_i| x_i^2 - (p^T v)^2 - (q^T v)^2 - \operatorname{tr}\big((KN)^T G(KN)\big). \tag{37}$$

This is the decisive nondegeneracy condition. The certificate verifies, by outward rational intervals,

$$H \succ \frac{3}{25} I_4, \qquad \sum_{i=1}^{4} \|K_i\|_F^2 < 100. \tag{38}$$

It follows from Cauchy–Schwarz that $\|K(x)\|_{\mathrm{op}} \le 10r$.

4.3.3 Second- and third-order corrections

Let $T_2(x) = \sum_{h=1}^{5} \beta_h(x) E_h$ be the real symmetric Toeplitz matrix with zero diagonal satisfying

$$N^T (T_2 - R_2) N = (KN)^T G(KN) + \tfrac{1}{2}(x^T H x) I_2. \tag{39}$$

We specify it uniquely by requiring β to be the minimum Euclidean-norm solution of the $(1,1)$ and $(1,2)$ entries of (39). Concretely, let

$$F = \begin{pmatrix} (p^T E_h p)_{h=1}^{5} \\ (p^T E_h q)_{h=1}^{5} \end{pmatrix}; \tag{40}$$

Define

$$D_2 = N^T R_2 N + (KN)^T G(KN) + \tfrac{1}{2}(x^T H x) I_2, \qquad d_2 = \big((D_2)_{11}, (D_2)_{12}\big)^T.$$

Then $\beta = F^T (F F^T)^{-1} d_2$; the certificate checks $F F^T \succ 0$. In particular, this definition uses only the already specified algebraic data, and no free choice of a second-order correction remains.

It is important that the third entry in the 2×2 symmetric equation is not an extra assumption. From (20), for every zero-diagonal real symmetric Toeplitz matrix U,

$$p^T U p + q^T U q = 0. \tag{41}$$

On the other hand, (37) and $s^T p = \Lambda$, $s^T q = 0$ show

$$\operatorname{tr}\left[N^T R_2 N + (KN)^T G (KN) + \tfrac{1}{2}(x^T H x) I_2\right] = 0. \tag{42}$$

Thus the $(2,2)$ entry follows from the other two, and (39) is consistent. Write

$$S_2(x) = T_2(x) - R_2(x) = \sum_{i,j=1}^{4} x_i x_j S_{ij}, \qquad S_{ij} = S_{ji}. \tag{43}$$

The program certifies

$$\sum_{i,j=1}^{4} \|S_{ij}\|_F^2 < 289, \qquad \text{hence} \qquad \|S_2(x)\|_{\mathrm{op}} \le 17 r^2. \tag{44}$$

For the cubic correction set $e_2 = -w_2/2$ and

$$R_3 = e_2 v^T - v e_2^T. \tag{45}$$

Let

$$J_* = \frac{\sum_{h=1}^{5} c_h J_h}{\sum_{h=1}^{5} c_h^2}, \qquad p^T J_* q = 1, \tag{46}$$

where c_h are as in (35); the denominator is positive by the invertibility checks. Define

$$\gamma_3(x) = p^T\big(R_3 + KGS_2 + S_2GK + KGKGK\big)q, \qquad A_3(x) = \gamma_3(x) J_*, \qquad K_3 = A_3 - R_3. \tag{47}$$

All matrix products are ordinary finite products. By skew-symmetry, the matrix

$$N^T\big(K_3 - KGS_2 - S_2GK - KGKGK\big)N \tag{48}$$

is real skew-symmetric; its only independent entry, the $(1,2)$ entry, vanishes by (47). Hence (48) is identically zero. The certificate obtains the homogeneous symmetric tensor $K_3(x) = \sum_{i,j,k} x_i x_j x_k C_{ijk}$ by polarization and verifies

$$\sum_{i,j,k=1}^{4} \|C_{ijk}\|_F^2 < 16, \qquad \text{hence} \qquad \|K_3(x)\|_{\mathrm{op}} \le 4 r^3. \tag{49}$$

4.3.4 Schur complement and explicit remainder

Let

$$T(x) = T_* + i A_1(x) + T_2(x) + i A_3(x). \tag{50}$$

Each correction is Hermitian Toeplitz (because i times a real skew-symmetric Toeplitz matrix is Hermitian), and each has zero diagonal, so $T(x)_{00} = \Lambda^2$.

For $b = s \odot e^{i\theta}$, the exact identities

$$\cos\theta_j - 1 = -\frac{x_j^2}{1 + \cos\theta_j}, \qquad \cos\theta_j - 1 + \frac{x_j^2}{2} = -\frac{x_j^4}{2(1+\cos\theta_j)^2}$$

show that

$$R_b = ss^T + iR_1 + R_2 + iR_3 + \Delta, \qquad \|\Delta\|_{\mathrm{op}} \le 4r^4 \tag{51}$$

for $r \le 1/125$. Here is an explicit remainder calculation. Set $u = -w_2/2$ and $e = s \odot (\cos\theta - \mathbf{1} + \widehat{x}^{\,2}/2)$, where $\cos\theta$ has entries 1 at indices $0, 5$. Since $\cos\theta_j > 1/2$ and $\sum_j x_j^8 \le (\sum_j x_j^2)^4 = r^8$, the second displayed identity gives

$$\|u\|_2 \le r^2/2, \qquad \|e\|_2 \le r^4/4, \qquad \|v\|_2 \le r, \qquad \|s\|_2 = \sqrt{6}.$$

Now $b = s + u + e + iv$ and $\overline{b} = s + u + e - iv$, so the contributions through total degree three are exactly those displayed in R_1, R_2, R_3. The remaining terms are

$$\Delta = uu^T + (s + u - iv)e^T + e(s + u + iv)^T + ee^T.$$

The rank-one operator-norm identity $\|ab^T\|_{\mathrm{op}} = \|a\|_2\|b\|_2$ therefore yields

$$\|\Delta\|_{\mathrm{op}} \le \frac{r^4}{4} + 2\big(\sqrt{6} + r + r^2/2\big)\frac{r^4}{4} + \frac{r^8}{16} < 4r^4 \qquad (0 < r \le 1/125).$$

At $r = 0$ the remainder is zero, so (51) holds throughout the stated ball.

Consequently

$$T(x) - R_b = S_0 + iK + S_2 + iK_3 - \Delta. \tag{52}$$

We analyze this Hermitian matrix relative to the orthogonal decomposition $\operatorname{ran} N \oplus (\operatorname{ran} N)^{\perp}$. Let

$$h(r) = 17r^2 + 4r^3 + 4r^4, \quad e(r) = 10r + h(r), \quad g(r) = 4r^3 + 4r^4. \tag{53}$$

By Lemma 4.2 and (38)–(51), the lower-right (range) block is bounded below by $(4 - e(r))I$. Its inverse has norm at most $1/(4 - e(r))$. Two applications of the resolvent identity, expanding around the inverse G of S_0 and retaining the constant and linear terms, bound the residual inverse by

$$\frac{h(r)}{16} + \frac{e(r)^2}{16(4 - e(r))}. \tag{54}$$

Indeed, if D is the range perturbation with $\|D\| \le e(r)$ and D_1 its linear term with $\|D - D_1\| \le h(r)$, the identity

$$(S_0 + D)^{-1} = G - GDG + GDGD(S_0 + D)^{-1}$$

on the range gives (54), using $\|G\| \le 1/4$ and replacing D by D_1 in the first-order term.

To make the Schur remainder exhaustive, write

$$A = iK, \quad S = S_2, \quad Z = iK_3 - \Delta, \quad B = A + S + Z, \quad V = S + Z.$$

Then $\|A\| \le 10r$, $\|Z\| \le g(r)$, $\|V\| \le h(r)$, and $\|B\| \le e(r)$. Let $\mathsf{P} = I_6 - \Pi$ be the orthogonal projection onto $\operatorname{ran}(N)^{\perp}$. The range-block perturbation is $D = \mathsf{P}B\mathsf{P}$ (restricted to that range), and its linear part is $D_1 = \mathsf{P}A\mathsf{P}$. With G viewed as zero on $\operatorname{ran}(N)$, write

$$(S_0|_{\operatorname{ran}\mathsf{P}} + D)^{-1} = G - GD_1G + \mathcal{R}, \qquad \|\mathcal{R}\| \le \frac{h(r)}{16} + \frac{e(r)^2}{16[4 - e(r)]}.$$

Indeed, twice applying the resolvent identity gives $G - GDG + GDGD(S_0 + D)^{-1}$, and replacing D by D_1 in the linear term introduces an error bounded by $h(r)/16$.

The Schur complement, expressed in the two-dimensional coordinate basis N (which need not be orthonormal), is

$$\mathcal{S} = N^T BN - N^T B(G - GD_1G + \mathcal{R})BN.$$

Because $N^T AN = 0$, its homogeneous terms of orders two and three are respectively

$$N^T SN - N^T AGAN = \tfrac{1}{2}(x^T Hx)I_2$$

and

$$N^T\big(iK_3 - AGS - SGA + AGAGA\big)N = iN^T\big(K_3 - KGS - SGK - KGKGK\big)N = 0.$$

We used $A = iK$ and (39), (48). The remaining difference has norm at most

$$E(r) = 4r^4 + \frac{20r\,g(r) + h(r)^2}{4} + \frac{10r}{16}\big(20r\,h(r) + h(r)^2\big) + [10r + h(r)]^2\left(\frac{h(r)}{16} + \frac{e(r)^2}{16[4 - e(r)]}\right). \tag{55}$$

For completeness, the four terms follow by estimating, in order, $\|N^T \Delta N\| \le 4r^4$;

$$\|BGB - AGA - AGS - SGA\| \le \frac{2(10r)g(r) + h(r)^2}{4};$$

$$\|BGD_1GB - AGD_1GA\| \le \frac{10r}{16}\big(2(10r)h(r) + h(r)^2\big);$$

and $\|B\mathcal{R}B\| \le e(r)^2\|\mathcal{R}\|$. Every displayed estimate uses only submultiplicativity, $\|G\| < 1/4$, $\|N\| \le 1$ and the preceding bounds. In these products G annihilates the nullspace, so the range projections may be suppressed without changing the expressions.

The certificate checks as rational inequalities, for $R = 1/125$,

$$e(R) < 4, \qquad E(R) < \frac{3}{50}R^2. \tag{56}$$

Every coefficient in $E(r)$ and $e(r)$ is nonnegative, and $4 - e(r)$ decreases on $[0, R]$; therefore $E(r)/r^2$ is nondecreasing there. Hence (56) applies to all $0 < r \le R$. By (38), the main Schur term is greater than $(3/50)r^2 I_2$, while its error is smaller than that. Both the range block and the Schur complement are strictly positive. Thus $T(x) - R_b \succ 0$ for $0 < r \le 1/125$; at $r = 0$ it equals $S_0 \succeq 0$.

Theorem 4.3 (Sharp local phase bound)**.** *For $\|\theta\|_2 \le 1/125$,*

$$L(1, s_1e^{i\theta_1}, \dots, s_4e^{i\theta_4}, 1) \le \Lambda, \tag{57}$$

with strict inequality unless $\theta = 0$. The same assertion, after the symmetries (23), holds in the corresponding balls around all ten phase centers (25).

Proof. Since $|\theta_j| \le 1/125 < \pi/2$, the equality $r = 0$ is equivalent to $\theta = 0$. Equations (50)–(56) give a Hermitian Toeplitz matrix with diagonal Λ^2 majorizing R_b, strictly when $\theta \ne 0$. Proposition 3.2 gives the desired bound. If $\theta \ne 0$, a norm-one polynomial attaining $L(b)$ exists by compactness, and strict positive definiteness makes its functional value strictly less than Λ. Symmetries preserve L. □

4.4 Excluding phase boxes without a global Lipschitz loss

Set

$$L_0 = \frac{21865297718925}{10^{13}} = 2.1865297718925 < \Lambda. \tag{58}$$

The strict comparison is proved by the rational enclosure for w in the certificate.

Lemma 4.4 (Phase-aligned box exclusion)**.** *Let $B(c,h)$ be a coordinate box in the normalized phase torus (24), centered at $c \in (\mathbb{R}/2\pi\mathbb{Z})^4$ and with each coordinate at angular distance at most $h < \pi/2$ from the corresponding coordinate of c. If*

$$L(b(c)) < L_0 \cos h, \tag{59}$$

then $B(c,h)$ cannot contain the phase vector associated with any maximizing degree-five polynomial.

Proof. A maximizing polynomial exists by compactness of the unit ball in the finite-dimensional coefficient space. For its coefficient vector a, choose $b_k = \overline{a_k}/|a_k|$ when $a_k \neq 0$ and any unit phase when $a_k = 0$. Normalize $b_0 = b_5 = 1$ using (23), with the corresponding norm-preserving transformation of the polynomial. Suppose the resulting phase vector lies in $B(c,h)$. With $\rho_k = |a_k|$, the center has coefficient phase errors $\delta_0 = \delta_5 = 0$ and $|\delta_k| \le h$ for $1 \le k \le 4$. Hence

$$L(b(c)) \ge \Re\Phi_{b(c)}(P) = \sum_{k=0}^{5} \rho_k \cos\delta_k \ge \cos h \sum_{k=0}^{5} \rho_k = \cos h\, S_5 \ge \cos h\, \Lambda > L(b(c)),$$

a contradiction. This statement concerns *globally maximizing aligned phases*; it does not assert $L(b(\theta)) \le L_0$ for every θ in the box. □

Two convenient sufficient tests for (59) follow from Proposition 3.2 and Fourier interpolation.

Lemma 4.5 (Five-point Fourier test)**.** *For $b = (1, b_1, b_2, b_3, b_4, 1)$, let $\zeta = e^{2\pi i/5}$ and*

$$d_j = \frac{1}{5}\sum_{k=0}^{4} b_k \zeta^{-jk} \quad (0 \le j \le 4). \tag{60}$$

Then $L(b) \le \sum_{j=0}^{4} |d_j|$.

Proof. By discrete Fourier inversion, $\sum_j d_j \zeta^{jk} = b_k$ for $k = 0, \dots, 4$. For $k = 5$, the left side equals $\sum_j d_j = b_0 = b_5 = 1$. Thus $\Phi_b(P) = \sum_j d_j P(\zeta^j)$, giving the bound. □

4.5 A complete finite covering of the phase torus

The certificate described here verifies a finite exact assertion. We specify every class of leaf, the recursive partition, the arithmetic model, the counts, and the connection to Lemma 4.4. In particular, the reader does not need to trust a numerical optimizer.

4.5.1 Initial cells and exact symmetry reduction

At level $\ell = 0$, take all boxes with centers

$$c = \frac{2\pi}{32}(t_1, t_2, t_3, t_4), \qquad t_j \in \{0, 1, \dots, 31\}, \qquad h_0 = \frac{\pi}{32}.$$

They cover the full normalized four-torus, including boundary points. Conjugation and reversal generate the transformations $t \mapsto -t$ and $t \mapsto (t_4, t_3, t_2, t_1)$ on this grid. The group has order four. Burnside's lemma gives

$$\frac{32^4 + 32^2 + 32^2 + 16}{4} = 262660 \tag{61}$$

representative boxes, selecting the lexicographically least tuple in each orbit. These symmetries preserve the problem and map each box to another box, so examining representatives suffices.

At level $\ell \geq 0$, put

$$h_\ell = \frac{\pi}{32 \cdot 2^\ell}, \qquad n_0 = 32, \quad n_\ell = 32 \cdot 2^{\ell+1} \ (\ell \geq 1), \tag{62}$$

and represent each center by an integer tuple t with $c = 2\pi t / n_\ell$. A nonterminal level-zero box has exactly sixteen children with center tuples $4t + (\pm 1, \pm 1, \pm 1, \pm 1)$ modulo 128. At all later levels the children have tuples $2t + (\pm 1, \pm 1, \pm 1, \pm 1)$ modulo $n_{\ell+1}$. This subdivision gives sixteen closed subboxes of half the original side length, with no gaps.

4.5.2 Four verified node types

Each visited box is assigned exactly one of the following four types.

1. **Fourier leaf.** Using (60) at the center, the certificate proves $\sum_j |d_j| < L_0 \cos h_\ell$ by outward interval arithmetic. Lemmas 4.5 and 4.4 exclude the entire box.

2. **Local leaf.** The certificate proves that the Euclidean angular distance from the box center to one of the ten points (25), plus $2h_\ell$ (the Euclidean half-diagonal of the box), is strictly less than $1/125$. The entire box is then contained in a ball treated by Theorem 4.3.

3. **Toeplitz leaf.** The certificate provides five complex rational moments $m_1, \ldots, m_5$, with real and imaginary parts of denominator 10^{12}, and defines the Hermitian Toeplitz matrix T by $T_{00} = (L_0 \cos h_\ell)^2$, $T_{j,k} = m_{k-j}$ for $j < k$, and conjugate symmetry. Outward interval LDL^* elimination proves $T - R_{b(c)} \succ 0$. Proposition 3.2 and Lemma 4.4 exclude the box.

4. **Subdivision node.** The program visits all sixteen children as specified above.

These alternatives are exhaustive: every node, starting from each of the (61) representatives, has exactly one recorded type; every subdivided node creates every one of its sixteen children. The verifier rejects missing or surplus records.

4.5.3 Certified counts and integer arithmetic

Proposition 4.6 (Finite phase-cover certificate)**.** *The accompanying degree-five finite verifier passes all tests described above with the following exact counts:*

Initial symmetry-orbit representatives	*262660*
All visited nodes	*272852*
Fourier leaves	*248376*
Local leaves	*549*
Toeplitz leaves	*23290*
Subdivision nodes	*637*
Maximum leaf level	*7*

The subdivision counts at levels $0, \dots, 6$ *are respectively* $94, 93, 101, 101, 101, 102, 45$. *All interval LDL* pivots are strictly positive; their smallest certified lower endpoint is*

$$\frac{3505258205231374557056651943921957879845929}{10^{50}} > 0.$$

The 2,136,052-byte uncompressed node-and-moment payload has SHA-256 digest

`5c9c8ad7091a626d37e0dfde3694d7e30c751f94c5ae52326c9e51180c9e1577`.

Proof. This is a finite exact computation. Appendix C specifies its decoding, arithmetic, coverage recursion, and verification conditions; `Sidon5_Exact_Closure_Certificate_Revised.py` implements them verbatim. The digest is an integrity check, not an alternative to the mathematical inequalities: all intervals, matrices, and child relations are recalculated and verified by the program. The recorded counts also satisfy $272852 = 262660 + 16 \cdot 637$, as required by exhaustive subdivision. □

Theorem 4.7 (Global phase exclusion)**.** *If* P *is a global maximizing polynomial for* S_5*, then its normalized aligned coefficient phases lie at one of the ten points* (25)*. Moreover* $S_5 = \Lambda$.

Proof. Normalize the aligned phase vector as in Lemma 4.4, and choose an initial grid box containing it. By symmetry, we may work with its representative. Proposition 4.6 recursively partitions this box until a verified leaf is reached. Fourier or Toeplitz leaves cannot contain the phase vector of a global maximizer, by Lemma 4.4; hence the phase lies in a local leaf, and so in the neighborhood of Theorem 4.3. Thus its phase-functional norm is at most Λ, and by alignment $S_5 = |\Phi_b(P)| \le L(b) \le \Lambda$. Combining with (22) proves equality. The strict part of Theorem 4.3 then forces the normalized phase to equal the corresponding center. □

4.6 Uniqueness and the full equality classification

It remains to determine the maximizing polynomial for a fixed optimal phase functional. By the symmetries (23), it is enough to treat $s = (1, 1, 1, 1, -1, 1)$.

Proposition 4.8 (Uniqueness at the sharp phase)**.** *If* $\|P\|_\infty \le 1$ *and* $\Phi_s(P) = \Lambda$*, then* $P = P_*$. *More generally, if* $|\Phi_s(P)| = \Lambda$*, then* P *is a unimodular multiple of* P_*.

Proof. The Toeplitz inequalities and $|\Phi_s(P)| = \Lambda$ yield

$$\Lambda^2 = |\Phi_s(P)|^2 \le a^* T_* a \le \Lambda^2 \|P\|_\infty^2 \le \Lambda^2.$$

If $\Phi_s(P) = \Lambda$, equality forces $a^* S_0 a = 0$, so Lemma 4.2 gives $a \in \ker_{\mathbb{C}} S_0 = \operatorname{span}_{\mathbb{C}}\{p, q\}$. Since $s^T p = \Lambda$ and $s^T q = 0$, the equation $s^T a = \Lambda$ implies

$$P = P_* + \beta Q_*, \qquad \beta \in \mathbb{C}. \tag{63}$$

We show $\beta = 0$ from the necessary angular derivative conditions at two contact points. By (21), at every zero z of Q_* on $\mathbb{T}$ we have $|P_*(z)| = 1$. Also $Q_*(1) = 0$ and $P_*(1) = 1$. Hence any feasible polynomial (63) has modulus one at $z = 1$, and its angular derivative of squared modulus must vanish there. As P_* has real coefficients,

$$0 = \frac{d}{dt}|P(e^{it})|^2\Big|_{t=0} = 2\,\mathrm{Re}\big(i(P_*'(1) + \beta Q_*'(1))\big).$$

Now $Q_*'(1) = (\kappa - 5)\sqrt{p_0 p_5} \in \mathbb{R} \setminus \{0\}$, forcing $\mathrm{Im}\,\beta = 0$.

The complementary polynomial factors as

$$Q_*(z) = \sqrt{p_0 p_5}\,(1-z)(1+z+\eta z^2+z^3+z^4). \tag{64}$$

The bound $0.834 < \eta < 0.835$ follows by rational isolation. For a unit-circle zero of the quartic, divide its equation by z^2 and set $x = \cos t$ to obtain $4x^2+2x+\eta-2=0$. Its two roots

$$c = \frac{-1+\sqrt{9-4\eta}}{4}, \qquad d = \frac{-1-\sqrt{9-4\eta}}{4} \tag{65}$$

are distinct and belong to $(-1,1)$; hence Q_* has five distinct unit-circle zeros, including 1. Choose $z = c + i\sqrt{1-c^2}$. The same derivative condition at this contact point, now with $\beta \in \mathbb{R}$, gives

$$0 = 2\beta\,\mathrm{Re}\left(\overline{P_*(z)}\,izQ_*'(z)\right). \tag{66}$$

The certificate evaluates the algebraic expression in parentheses with rigorous rational intervals and proves it is nonzero. Thus $\beta = 0$. For arbitrary complex value $\Phi_s(P)$ of modulus Λ, multiply P by a scalar of modulus one to make this value Λ and apply the result. □

Proof of the degree-five upper bound and extremizer statement. Section 4.1 gives the attainable lower bound Λ. Theorem 4.7 supplies the matching global upper bound and restricts equality phases to (25). Proposition 4.8 classifies the norming polynomial at one center. Undoing scalar and variable rotations and coefficient reversal gives exactly (17). Conversely, every polynomial in (17) has supremum norm one and coefficient modulus sum Λ, by (21). The real polynomial P_* attains the complex optimum, so the real and complex Sidon constants agree. □

5 Degrees six and seven: unrestricted global optimality

5.1 Stationary root boxes, feasibility and lower bounds

We now specify the candidates without assigning a numerical approximation the status of an exact algebraic definition. Fix $n \in \{6,7\}$ and write $a_k = x_k + iy_k$, taking the gauge $y_0 = y_1 = 0$ and $x_0, x_1 > 0$. Introduce n putative contacts $z_j = u_j + iv_j$, positive weights λ_j, and positive amplitudes ρ_k. Let

$$P(z) = \sum_{k=0}^{n} a_k z^k, \quad DP(z) = zP'(z), \quad f_k = \rho_k \sum_{j=1}^{n} \lambda_j \overline{P(z_j)} z_j^k - \overline{a_k}.$$

The following system, when split into real and imaginary parts, consists of $6n+1$ *rational real polynomial equations* in the same number of real variables:

$$\begin{aligned} |z_j|^2 = 1, \quad |P(z_j)|^2 = 1, \quad \mathrm{Im}\big(\overline{P(z_j)}DP(z_j)\big) = 0 \quad & (1 \le j \le n), \\ \rho_k^2 = |a_k|^2, \qquad \mathrm{Re}\, f_k = 0 \quad & (0 \le k \le n), \\ \mathrm{Im}\, f_k = 0 \quad & (2 \le k \le n). \end{aligned} \tag{67}$$

The order of variables is

$$(x_0,\dots,x_n);\ (y_2,\dots,y_n);\ (u_1,\dots,u_n);\ (v_1,\dots,v_n);\ (\lambda_1,\dots,\lambda_n);\ (\rho_0,\dots,\rho_n). \tag{68}$$

For each n, Appendix A specifies a rational box $\mathcal{B}_n$ by giving all centers with denominator 10^{15} and common radius 10^{-12}. The finite verifiers check that (67) has exactly one real solution in $\mathcal{B}_n$, that

its weights and amplitudes are positive and its contacts distinct, and that $x_0, x_1 > 0$. Denote that solution by ξ_n, its polynomial by P_n^*, and set

$$C_n = \sum_{k=0}^{n} \rho_k(\xi_n) \quad (n = 6, 7). \tag{69}$$

Since the system and isolating boxes have rational data, C_n is a uniquely defined real algebraic number. At this stage this is a multivariate algebraic specification. Section 7 upgrades it to the one-variable minimal-polynomial descriptions in Theorems 1.2 and 1.3.

Lemma 5.1 (Isolation and feasibility interface)**.** *Each $\mathcal{B}_n$ contains exactly one real solution of (67). The corresponding P_n^* satisfies $\|P_n^*\|_{\mathbb{T}} = 1$, has $n+1$ nonzero coefficients and precisely n distinct, nondegenerate contacts, and*

$$\mathcal{J}(P_n^*) = C_n, \qquad \sum_{j=1}^{n} \lambda_j = C_n, \qquad \sum_j \lambda_j \overline{P_n^*(z_j)} z_j^k = \frac{\overline{a_k}}{\rho_k} \quad (0 \le k \le n).$$

Proof. We specify the angular system used for the finite existence and uniqueness test. Its $4n$ coordinates are $u = (x_0, \dots, x_n, y_2, \dots, y_n, t_1, \dots, t_n, \lambda_1, \dots, \lambda_n)$, with $z_j = e^{it_j}$ and $\rho_k = |a_k| > 0$. Define

$$r_k = \sum_j \lambda_j \overline{P(e^{it_j})} e^{ikt_j} - \overline{a_k}/|a_k|.$$

In coordinate order the real map F consists of $\operatorname{Re} r_0, \dots, \operatorname{Re} r_n$, $\operatorname{Im} r_2, \dots, \operatorname{Im} r_n$, the n values $|P(e^{it_j})|^2 - 1$, and the n values $-\operatorname{Im}(\overline{P(e^{it_j})} DP(e^{it_j}))$. The angular box B has rational center c (denominator 10^{15}) and independent radius 10^{-9}, with center and rational preconditioner R supplied in the certificate. This is the angular uniqueness box; the Cartesian box $\mathcal{B}_n$ in the appendix has radius 10^{-12}. Outward interval automatic differentiation checks

$$\sup_{u \in B} \|I - RDF(u)\|_\infty < \tfrac{1}{2}, \qquad c - RF(c) + (I - RDF(B))(B - c) \subset \operatorname{int} B. \tag{70}$$

Thus $u \mapsto u - RF(u)$ contracts B into itself. Its unique fixed point is a zero of F, because the first inequality also implies that R is nonsingular. The code then certifies that this root maps into the full rational box $\mathcal{B}_n$, and that every solution in $\mathcal{B}_n$ maps back into B; on a sufficiently short arc, a chord displacement d bounds angular displacement by $2d$. Positivity of ρ_k identifies the positive square root uniquely. Therefore the two box representations describe the same unique solution.

Two imaginary equations were omitted in (67). For the residuals r_k just defined, the listed equations imply that only $\operatorname{Im} r_0, \operatorname{Im} r_1$ could remain. Since $|P(z_j)| = 1$,

$$\sum_k a_k r_k = \sum_j \lambda_j - \sum_k \rho_k \in \mathbb{R},$$

and, by the tangency equations, $\operatorname{Im} \sum_k k a_k r_k = 0$. As a_0, a_1 are strictly positive real, the second identity forces $\operatorname{Im} r_1 = 0$, then the first forces $\operatorname{Im} r_0 = 0$. Therefore all complex moment equations hold, and their contraction against a_k yields $\sum_j \lambda_j = C_n$.

For complete circle feasibility, put $V(z) = \prod_{j=1}^{n} (z - z_j)$. Nonnegativity of the candidate defect is *not* assumed beforehand. Instead, the value and tangency equations show that the Laurent polynomial $1 - |P|^2$ has a double zero at each of the n distinct contacts. Multiplication by z^n produces a polynomial of degree at most $2n$; hence it is proportional to $V(z)\, z^n \overline{V(1/\bar{z})}$, equivalently

$$1 - |P(e^{it})|^2 = \alpha\, |V(e^{it})|^2, \quad \alpha = \frac{1 - \sum_k |a_k|^2}{\sum_k |[z^k]V|^2}. \tag{71}$$

The verifiers check $\sum |a_k|^2 < 1$, so $\alpha > 0$. Equation (71) proves $\|P\|_{\mathbb{T}} \le 1$ globally, with equality precisely at the n listed nodes and a strict quadratic loss away from each simple zero of V. Positivity of the amplitudes, weights, and pairwise distinctness of contacts are separately certified interval inequalities. All asserted properties follow. □

The lemma supplies an important general lesson: a solution of the critical equations is not automatically feasible. Feasibility here follows from a *global polynomial identity* with a rigorously positive coefficient, not from inspecting the circle at finitely many approximate points.

5.2 A locally sharp certificate with a one-dimensional real kernel

Let $n \in \{6, 7\}$, and abbreviate $C = C_n$, $P = P_n^*$, $\rho = (\rho_0, \dots, \rho_n)^T$, $b_k = \bar{a}_k/\rho_k$ and $z_j = e^{it_j}$. At the center $\nu = 0$ we have a positive atomic representation with $\sum \lambda_j = C$. Set $d = n - 1$. We use the end-point gauge $\theta_0 = \theta_n = 0$ for the varying phases, with a *rational invertible preconditioner* $B \in \mathbb{Q}^{d\times d}$ and $\theta = B\nu$. For $k = 0, n$ let the kth row Φ_k of the extended phase matrix vanish; the interior rows are those of B.

Fix rational matrices $A, L \in \mathbb{Q}^{n\times d}$ with $\sum_j L_{jh} = 0$ for each h. Move the contact angles and positive weights by

$$t_j(\nu) = t_j + (A\nu)_j, \qquad \lambda_j(\nu) = \lambda_j + (L\nu)_j.$$

Define a positive Toeplitz matrix $T(\nu)$ by the atomic measure $C\sum_j \lambda_j(\nu)\delta_{e^{it_j(\nu)}}$, and put

$$S(\nu) = \mathrm{Re}\big(D_{b\odot e^{iB\nu}} T(\nu) D_{\overline{b\odot e^{iB\nu}}}\big) - \mathbf{1}\mathbf{1}^T. \tag{72}$$

The endpoint phases in this formula remain fixed. Whenever $\lambda_j(\nu) > 0$ and $S(\nu) \succeq 0$, Theorem 3.3 proves $M_n(b\odot e^{iB\nu}) \le C$. Notice that $T(\nu)$ is *automatically positive*, a substantial economy compared with directly correcting a singular complex Toeplitz matrix.

Lemma 5.2 (Finite Taylor–Schur criterion)**.** *Assume $\|\rho\|_2 \le 1$, $S(0)\rho = 0$ and $\ker_{\mathbb{R}} S(0) = \mathbb{R}\rho$. The norm condition holds for the candidate by Parseval (indeed the certificate checks $\sum_k |a_k|^2 < 1$). Let $W \in \mathbb{Q}^{m\times n}$ be such that $[\rho\ W]$ is invertible and $W^T S(0) W \succeq \tau I_n$ for some $\tau > 0$. Write $S(\nu) = S_0 + S_1(\nu) + S_2(\nu) + S_3(\nu) + \Delta_4(\nu)$, where S_j is homogeneous of degree j. Suppose $\rho^T S_1(\nu)\rho = 0$. Define*

$$h(\nu) = -W(W^T S_0 W)^{-1} W^T S_1(\nu)\rho, \quad H[\nu] = \rho^T S_2(\nu)\rho - (W^T S_1(\nu)\rho)^T (W^T S_0 W)^{-1} (W^T S_1(\nu)\rho).$$

Assume $H[\nu] \ge \mu\|\nu\|^2$ with $\mu > 0$, and the tensor norm estimates and scalar endpoint inequalities (74)–(76) below. Then $S(\nu) \succ 0$ whenever $0 < \|\nu\| \le R$, while $S(0) \succeq 0$ with kernel $\mathbb{R}\rho$. If also $\lambda_j > R\|L_j\|_2$ for each row, this gives a sharp local bound $M_n \le C$, strict away from the center.

Proof. We give the exact finite formulae, so the certificate's checked quantities have a defined mathematical meaning. Put $w_{jk} = \bar{b}_k e^{ikt_j}$ and $\beta_{j,ik,h} = (k - i)A_{jh} - (\Phi_k - \Phi_i)_h$. Then the (i, k) entry of $S(\nu)$ is

$$C\sum_j (\lambda_j + L_j\nu)\,\mathrm{Re}\left(\bar{w}_{ji} w_{jk} \exp\big(i\sum_h \beta_{j,ik,h}\nu_h\big)\right) - 1.$$

The homogeneous tensor of order $q \in \{1, 2, 3\}$, for an ordered multi-index $(h_1, \dots, h_q)$, has (i, k) entry

$$\frac{C}{q!}\sum_j \left[\lambda_j\,\mathrm{Re}(i^q \bar{w}_{ji} w_{jk}) \prod_{r=1}^{q} \beta_{j,ik,h_r} + \sum_{r=1}^{q} L_{j,h_r}\,\mathrm{Re}(i^{q-1}\bar{w}_{ji} w_{jk}) \prod_{s\ne r} \beta_{j,ik,h_s}\right]. \tag{73}$$

The scalar exponential Taylor bound $|e^{iu} - \sum_{q=0}^{3}(iu)^q/q!| \le |u|^4/24$ and its order-two counterpart for the terms multiplied by $L_j\nu$ yield $\|\Delta_4\| \le e_4 r^4$ for $r = \|\nu\|$, where the verifier forms e_4 from the Frobenius norm of the entrywise upper bounds

$$C \sum_j \left(\lambda_j \|\beta_{j,ik}\|^4/24 + \|L_j\| \|\beta_{j,ik}\|^3/6\right).$$

Positive weights and stationarity yield $\rho^T S_1 \rho = 0$: the derivative of the atomic quadratic form in the node variables vanishes at contacts; the weight derivatives sum to zero; the phase derivative vanishes by the complex moment equations.

Set $r = \|\nu\|$. Since $h(\nu) \in \operatorname{ran} W$, the matrix $[\rho + h(\nu), W]$ is invertible whenever $[\rho, W]$ is. In this basis $\|\rho + h(\nu)\|_2 \le 1 + ar$, which justifies the remainder factors used below. The change of basis cancels the first-order off-diagonal block because $W^T S_0 h + W^T S_1 \rho = 0$. In this basis, its scalar block has quadratic part $H[\nu]$ and cubic part

$$\Gamma_3 = \rho^T S_3 \rho + 2h^T S_2 \rho + h^T S_1 h,$$

and the off-diagonal block starts at order two, with terms $W^T(S_2\rho + S_1 h)$ and $W^T(S_3\rho + S_2 h)$. These are identities obtained by multiplying the homogeneous expansions, not numerical asymptotics.

Assume, for $r \le R$, that the following exact rational bounds have been certified:

$$\begin{gathered} \|h\| \le ar, \quad |\Gamma_3| \le g_3 r^3, \quad \|W^T(S_2\rho + S_1 h)\| \le c_2 r^2, \\ \|W^T(S_3\rho + S_2 h)\| \le c_3 r^3, \quad \|S_2\| \le s_2 r^2, \quad \|S_3\| \le s_3 r^3, \\ \|W^T S_j W\| \le k_j r^j \quad (j = 1, 2, 3), \quad \|W\| \le w, \quad \|\Delta_4\| \le e_4 r^4. \end{gathered} \tag{74}$$

For $0 < r \le R$ put

$$\begin{aligned} e(r) &= k_1 r + k_2 r^2 + k_3 r^3 + w^2 e_4 r^4, \\ A(r) &= \mu r^2 - g_3 r^3 - \left(2as_3 + a^2 s_2 + a^2 s_3 r + e_4(1 + ar)^2\right) r^4, \\ D(r) &= c_2 r^2 + c_3 r^3 + w s_3 a r^4 + w e_4 (1 + ar) r^4. \end{aligned} \tag{75}$$

The range block is at least $(\tau - e(r))I$, the scalar block at least $A(r)$, and the off-diagonal block has norm at most $D(r)$. Thus the Schur complement is strictly positive whenever

$$e(R) < \tau, \qquad A(R) > \frac{D(R)^2}{\tau - e(R)}. \tag{76}$$

After division by r^2, every error term in these inequalities is nonnegative and nondecreasing with r, while the positive quadratic main term is constant. Therefore checking R implies the result for all $0 < r \le R$. The fixed positive weights persist by $\lambda_j > R\|L_j\|_2$. Strict matrix positivity yields a strict amplitude inequality for all nonzero polynomials, completing the proof. □

For $n = 6, 7$ the certificate checks the following rational constants; its corresponding verified ellipsoid and guaranteed Euclidean phase radii appear below.

Parameter	$n=6$	$n=7$
$\tau=\mu$	99/100	99/100
(k_1,k_2,k_3)	$(6,19,9)$	$(5,10,10)$
(s_2,s_3)	$(21,13)$	$(48,53)$
a	27/100	21/20
g_3	17/10	8
(c_2,c_3)	$(6,3)$	$(8,9)$
e_4	48	380
w	17/10	89/100
R	1/16	1/32
Euclidean phase radius	1/64	1/128

The matrices A, L, B, W are printed in Appendix B as exact rational data and embedded in the verifier. For each n it reconstructs (73) and checks all matrix inequalities and scalar comparisons, rather than trusting these printed bounds as independent axioms. The condition $\|B^{-1}\| < 4$ guarantees the stated Euclidean radii.

5.3 The global phase certificate and exhaustive coverage

The transformations $b_k \mapsto \gamma\omega^k b_k$, conjugation and coefficient reversal preserve both L_n and M_n. Normalize $b_0 = b_n = 1$, leaving $d = n-1$ internal phases. The residual rotations $\omega^n = 1$, phase negation and internal-coordinate reversal are exact symmetries of this normalized torus.

5.3.1 Alignment excludes whole boxes

For an aligned feasible polynomial $a_k = \bar{b}_k\rho_k$ with $\rho_k \geq 0$, suppose its normalized phase lies in a box of center c and coordinate half-widths $h_k < \pi/2$. Endpoint errors are zero, so

$$\operatorname{Re}\Phi_{b(c)}(P) = \sum_{k=0}^{n} \rho_k \cos\delta_k \geq \cos(\max h_k)\mathcal{J}(P). \tag{77}$$

Thus a strict center estimate $L_n(b(c)) < C_n\cos(\max h_k)$ excludes every aligned feasible polynomial with $\mathcal{J}(P) \geq C_n$ from the entire box. As in degree five, this is a statement about aligned competitors, not a sharp pointwise bound at every phase in the box.

5.3.2 The four leaf predicates

The verifier rechecks each of the following predicates, rather than accepting the stored leaf label.

1. *DFT exclusion.* For $\zeta = e^{2\pi i/n}$ and $b_0 = b_n = 1$,

$$d_j = \frac{1}{n}\sum_{k=0}^{n-1} b_k\zeta^{-jk}, \quad \Phi_b(P) = \sum_{j=0}^{n-1} d_j P(\zeta^j), \quad L_n(b) \leq \sum_j |d_j|.$$

   Compare this center bound strictly with the aligned threshold.

2. *Interpolation exclusion.* For $n+1$ distinct roots of unity ζ_j, let $V(z) = \prod_j(z-\zeta_j)$ and $\ell_j(z) = V(z)/((z-\zeta_j)V'(\zeta_j)) = \sum_k \ell_{jk}z^k$. Lagrange interpolation gives $L_n(b) \leq \sum_j |\sum_k b_k\ell_{jk}|$. Each atlas entry records integer orders and exponents of roots of unity; its interpolation coefficients and strict comparison are reconstructed.

3. *Toeplitz exclusion.* Rebuild a rational Hermitian Toeplitz T with $T - \bar{b}b^T \succ 0$ and diagonal at most the threshold squared. Every pivot in interval complex LDL^* must have positive lower endpoint. Lemma 3.1 gives $L_n(b) \le \sqrt{T_{00}}$, with strictness at a nonzero norming polynomial because the matrix difference is positive definite. Compactness then gives the required strict center estimate.

4. *Local sharpness.* Certify that the entire box is contained in a symmetry image of a proved local ellipsoid. With the rational phase preconditioner B and half-width vector h, it suffices that

$$\|B^{-1}\delta_{\text{center}}\|_2 + \max_{s\in\{\pm1\}^d} \|B^{-1}\operatorname{diag}(h)s\|_2 < R. \tag{78}$$

   The maximum of this convex norm on the width box is attained at a vertex. Circular displacements, including wrap-around, are rigorously bounded. Lemma 5.2 gives $M_n \le C_n$ throughout this leaf and strictness except at a candidate center.

5.3.3 Initial orbits, branching and exhaustion

Start with centers $2\pi t/(2n)$ and half-width $\pi/(2n)$ in every internal coordinate. Reduce the $(2n)^{n-1}$ cells by the residual rotations, negation and reversal using lexicographically least integer representatives. There are 10616 representatives at degree six and 269712 at degree seven. Every branch splits *all* $n-1$ coordinates and has exactly 2^{n-1} children, with their common boundaries included. Integer centers and levels determine every child. The complete depth-first traversal has these verified statistics:

Quantity	$n=6$	$n=7$
Initial orbit representatives	10616	269712
DFT exclusion leaves	8653	220856
Interpolation exclusion leaves	15418	221505
Toeplitz exclusion leaves	37	0
Local sharp leaves	985	12949
Branch nodes	467	2946
All nodes	25560	458256
Maximum refinement depth	7	7

In both columns, the node count equals the initial count plus 2^{n-1} times the branch count. These identities are consistency checks; proof also requires every leaf inequality, every child relation, the enumeration of all initial orbits and exact payload exhaustion.

Theorem 5.3 (Unrestricted global upper bounds)**.** *The finite predicates above imply $S_6 = C_6$ and $S_7 = C_7$ for complex coefficients. Equality phases are symmetry images of the candidate phases.*

Proof. A norm-one global maximizer exists. Align its coefficient phases, using arbitrary phases at zero coefficients, normalize the endpoint phases, and take an initial orbit representative containing it. Traverse its complete verified tree to a containing leaf. A strict DFT, interpolation or Toeplitz leaf contradicts (77), since the feasible candidate gives $S_n \ge C_n$. At a local leaf, alignment and real duality give $\mathcal{J}(P) \le M_n(b) \le C_n$. Hence $S_n = C_n$. Strict local matrix positivity away from the center forces an equality phase to be a candidate phase. This argument covers zero coefficients and deficient contact competitors without assuming their absence. □

5.4 Rigidity and the reversal identity used in elimination

At a candidate phase the real matrix $S(0)$ has kernel $\mathbb{R}\rho$. If an aligned norm-one polynomial with amplitude vector $x \geq 0$ attains C_n, then

$$0 \leq x^T S(0)x = C_n \sum_j \lambda_j |P(z_j)|^2 - (\mathbf{1}^T x)^2 \leq 0.$$

Thus $x \in \mathbb{R}\rho$, and $\mathbf{1}^T x = C_n$ gives $x = \rho$. Consequently all normalized extremizers are scalar and variable rotations of P_n^* or its coefficientwise conjugate. Reversal introduces no new orbit: after normalization and contact reordering the reversed stationary point returns strictly to the angular box B of radius 10^{-9} in Lemma 5.1. Uniqueness in that box identifies the two points. The gauge $a_0, a_1 > 0$ is unique, while $\operatorname{Im} a_2 \neq 0$ by its certified interval; conjugation therefore gives a distinct orbit. These observations also justify the reciprocal representative used in the next section.

6 Rational charts and six-variable critical systems

The global phase certificates in Sections 4 and 5 prove optimality without eliminating their algebraic parameters. We now perform the separate identification needed for a minimal polynomial. Crucially, the symmetry used in this section is *proved for the globally certified candidate*, and is not assumed for all polynomials in the definition of S_n.

6.1 A reciprocal representative and its real complement

Let a_k denote the coefficients of the certified candidate in the gauge $a_0, a_1 > 0$. Set

$$\gamma = \frac{\overline{a_n}}{|a_n|}, \qquad \eta = \frac{\overline{\gamma a_{n-1}}}{|a_{n-1}|}, \qquad \widetilde{a}_k = \gamma \eta^k a_{n-k}.$$

These choices make $\widetilde{a}_0, \widetilde{a}_1$ positive real. The transformed contacts are $\widetilde{z}_j = (\eta z_j)^{-1}$, with the same positive weights, reordered by angle. Substitution in the moment, value and tangency equations proves that this is another stationary solution. The verifier encloses its angular coordinates strictly inside B, the radius-10^{-9} uniqueness box of Lemma 5.1. Its uniqueness gives $a_k = \gamma \eta^k a_{n-k}$ exactly. Applying this identity twice gives $\gamma^2 \eta^n = 1$, since $a_0 \neq 0$.

Choose $\zeta \in \mathbb{T}$ with $\zeta^2 = \eta^{-1}$ and put $P(z) = P_n^*(\zeta z)/a_0$. Its coefficients satisfy $P = \sigma P^{\mathrm{rev}}$ with $\sigma = \gamma \zeta^{-n}$ and $\sigma^2 = 1$. If n is even, the nonzero middle coefficient forces $\sigma = 1$. If n is odd, replacing ζ by $-\zeta$ changes the sign of σ, so choose the positive sign. Thus P is palindromic with $P_0 = P_n = 1$. The complementary factor constructed below then gives

$$P = P^{\mathrm{rev}}, \quad P_0 = P_n = 1, \quad |P|^2 + |Q|^2 = E \quad \text{on } \mathbb{T}, \qquad Q \in \mathbb{R}[z],\ Q = -Q^{\mathrm{rev}},\ Q_0 = 1,\ Q_n = -1. \tag{79}$$

Here $E = a_0^{-2} > 0$, since the original candidate has norm one; thus normalization to norm one divides P by $\sqrt{E}$. We justify all assertions about Q. Palindromicity makes $|P(e^{it})|$ even in t, so its full set of n contacts is closed under conjugation. The monic product of their linear factors is consequently real. Normalize that product to have constant coefficient one. All its roots are on $\mathbb{T}$, so it is real reciprocal or real antireciprocal, with leading coefficient $+1$ or -1. The highest autocorrelation in a complementary identity, with $P_0 = P_n = 1$, forces the latter sign and forces the squared scale of this normalized complement to be one. Thus (79) follows from full contact and the exact feasibility identity; enclosing an imaginary part in an interval containing zero would not suffice.

6.2 The degree-six rational chart

For real b, c, t, put $B = b^2$ and define the palindromic polynomial by

$$\begin{aligned} P_0 &= P_6 = 1, \\ P_1 &= P_5 = t(2c - b) + ib, \\ P_2 &= P_4 = (bc - B/4)(1 - t^2) - t^2 + it(2 + 2bc - B/2), \\ P_3 &= -bt(bc + 2) + i\{c(B - 4)(1 - t^2)/2 - 2bt^2\}. \end{aligned} \tag{80}$$

Set $q_1 = \operatorname{Re} P_1$, $q_2 = \operatorname{Re} P_2 + B/2$ and

$$Q = (1, q_1, q_2, 0, -q_2, -q_1, -1), \qquad E_6 = \frac{(B + 4)^2(1 + t^2)^2(1 + c^2)}{4}. \tag{81}$$

Direct autocorrelation expansion gives $|P|^2 + |Q|^2 = E_6$ on $\mathbb{T}$. These are exact polynomial identities, checked symbolically by `chart6_geometry.identities` as well as justified by substitution.

This chart covers a neighborhood of the normalized global candidate. Indeed, write $a = \operatorname{Re} P_1$, $b = \operatorname{Im} P_1$, $e = \operatorname{Im} P_2$. For $bt \neq 0$ its inverse is

$$t = \frac{2(e - ab)}{b^2 + 4}, \qquad c = \frac{a + bt}{2t}. \tag{82}$$

To verify local exhaustiveness rather than only substitution in one direction, the highest three nonconstant autocorrelations in an arbitrary pair (79) give $q_1 = a$, $q_2 = \operatorname{Re} P_2 + b^2/2$ and $\operatorname{Re} P_3 = ab^2/2 - be$. The remaining two independent lag equations are linear in $\operatorname{Re} P_2, \operatorname{Im} P_3$. Their determinant is $(b^2 + 4)^2 t/2 \neq 0$, and their solution is precisely (80). The certificate checks the lag equations and this determinant identity exactly, and checks $bt \neq 0$ at the global point.

Use the sign invariants

$$B = b^2, \qquad C = bc, \qquad U = t^2,$$

whose Jacobian determinant with respect to (b, c, t) is $4b^2 t \neq 0$. For $r_j = |P_j| > 0$, $1 \leq j \leq 3$, their squared values and energy are

$$\begin{aligned} R_1 &= B + U(2C - B)^2/B, \\ R_2 &= \{(C - B/4)(1 - U) - U\}^2 + U(2 + 2C - B/2)^2, \\ R_3 &= BU(C + 2)^2 + \{C(B - 4)(1 - U) - 4BU\}^2/(4B), \\ E_6 &= (B + 4)^2(1 + U)^2(B + C^2)/(4B). \end{aligned} \tag{83}$$

Thus the objective is $(2 + 2r_1 + 2r_2 + r_3)/\sqrt{E_6}$.

6.3 A uniform symmetric product and its rational inverse

For a polynomial of degree at most d, use the fixed-length reversal $\operatorname{rev}_d f(z) = z^d f(1/z)$, padding its coefficient list with zeros to length $d + 1$. In this subsection, reversal always uses the stated ambient degree, even when a leading coefficient vanishes. For a polynomial of degree at most d, use the fixed-length reversal $\operatorname{rev}_d f(z) = z^d f(1/z)$, padding its coefficient list with zeros to length $d + 1$. In this subsection, reversal always uses the stated ambient degree, even when a leading coefficient vanishes. For any $n \geq 2$, let $m = \lfloor n/2 \rfloor$, take real parameters $t_1, \ldots, t_m$, and set

$$\begin{aligned} W(z) &= \frac{1}{2}\begin{pmatrix} 1 + z & 1 - z \\ 1 - z & 1 + z \end{pmatrix}, \qquad D(t) = \operatorname{diag}(1 + it, 1 - it), \\ \mathcal{U}_n &= 2WD(t_1)WD(t_2)\cdots D(t_{n-1})W, \qquad t_j = t_{n-j}. \end{aligned} \tag{84}$$

The factors are arranged in the displayed symmetric order, using the independent $t_1, \dots, t_m$ when indices exceed m. The first row is (P, Q). Since W is unitary on $\mathbb{T}$ and $D(t)D(t)^* = (1+t^2)I$, it satisfies (79), with

$$E_n = 4\prod_{j=1}^{m}(1+t_j^2)^{e_j}, \qquad e_j = 2 \quad \text{except } e_m = 1 \text{ when } n \text{ is even.} \tag{85}$$

For clarity, transpose symmetry of the product gives its symmetric off-diagonal entries. Conjugation of the coefficient matrix is its conjugation by the exchange matrix, which interchanges the diagonal entries. These two facts make Q real. Replacing z by $1/z$ gives $P^{\mathrm{rev}} = P$, $Q^{\mathrm{rev}} = -Q$, and evaluation of the constant coefficient gives $P_0 = Q_0 = 1$. Alternatively, each assertion follows by induction on the factors. These identities are also symbolically audited.

The inverse is essential for identifying an arbitrary certified symmetric point with this chart. Write $P = A + iB_0$ with real palindromic A, B_0. Then

$$A^2 + B_0^2 - Q^2 = E_n z^n. \tag{86}$$

For $n \geq 3$, put $t = [z]B_0/2$ and

$$a = A + Q, \quad b = B_0/z, \quad c = (A-Q)/z^2, \quad d = (1+t^2)^2,$$

and form

$$a' = (a + 2tb - t^2 c)/d, \quad c' = (c + 2tb - t^2 a)/d, \quad b' = \{(1-t^2)b - t(a+c)\}/d. \tag{87}$$

The reduced pair is $P' = (a'+c')/2 + ib'$, $Q' = (a'-c')/2$, of degree at most $n-2$. To check that every recursive step is legitimate, the coefficients of z, z^2 in (86) give $A_1 = Q_1$ and $A_2 - Q_2 = -B_{0,1}^2/2 = -2t^2$. Together with the endpoint coefficients, these identities show that a, b, c are polynomials of degree at most $n-2$, with $a_0 = 2, b_0 = 2t, c_0 = -2t^2$. Hence (87) gives $a'_0 = 2, c'_0 = b'_0 = 0$. With ambient degree $n-2$, reversal exchanges a, c and fixes b; it therefore exchanges a', c' and fixes b'. Consequently

$$P'_0 = P'_{n-2} = Q'_0 = 1, \qquad Q'_{n-2} = -1, \quad \mathrm{rev}_{n-2}\, P' = P', \quad \mathrm{rev}_{n-2}\, Q' = -Q', \quad Q' \in \mathbb{R}[z].$$

Direct multiplication also gives

$$a'c' + (b')^2 = \frac{ac + b^2}{(1+t^2)^2} = \frac{E_n}{(1+t^2)^2} z^{n-2}.$$

Thus $|P'|^2 + |Q'|^2 = E'$ on $\mathbb{T}$, with $E' = E_n/(1+t^2)^2 > 0$, and the reduced pair has all the original normalization properties.

For the reverse step, (87) is inverted by

$$a = a' - 2tb' - t^2 c', \qquad c = c' - 2tb' - t^2 a', \qquad b = t(a'+c') + (1-t^2)b'.$$

Writing coefficientwise conjugation as a bar, these formulas are equivalent to the matrix identity

$$\begin{pmatrix} P & Q \\ Q & \bar{P} \end{pmatrix} = W D(t) \begin{pmatrix} P' & Q' \\ Q' & \overline{P'} \end{pmatrix} D(t) W.$$

It proves exact recovery of the two outer equal factors. For degree one the normalized pair is necessarily $(1+z, 1-z)$, of energy 4. For degree two, antireversal gives $Q_1 = 0$ and the lag-one

equation gives $\operatorname{Re} P_1 = 0$, so the pair is $(1 + 2itz + z^2, 1 - z^2)$, of energy $4(1 + t^2)$. Induction now proves the product representation and its rational inverse, whose only step denominator is $(1 + t^2)^2 > 0$. No symmetry of an arbitrary Sidon extremizer is assumed here.

For degree seven there are three parameters. Simultaneous coefficient conjugation changes (t_1, t_2, t_3) to its negative, so use

$$B = t_1^2, \quad C = t_1 t_2, \quad D = t_1 t_3, \qquad \det \frac{\partial(B, C, D)}{\partial(t_1, t_2, t_3)} = 2t_1^3. \tag{88}$$

Let $P_j(t)$ denote the explicitly defined coefficients of (84). Define $R_j(B, C, D)$ as the invariant rational expression $|P_j(t)|^2$, $1 \le j \le 3$, under (88). This is an explicit finite definition: expand the seven W factors, substitute $t_1^2 = B, t_2 = C/t_1, t_3 = D/t_1$, and replace every even power of t_1 by the corresponding power of B. Invariance ensures that no odd power remains. Then

$$E_7 = \frac{4(1 + B)^2(B + C^2)^2(B + D^2)^2}{B^4}, \qquad \frac{\mathcal{J}(P)}{\sqrt{E_7}} = \frac{(1 + r_1 + r_2 + r_3)B^2}{(1 + B)(B + C^2)(B + D^2)}, \quad r_j = \sqrt{R_j} > 0. \tag{89}$$

For the positive real branch, all denominators in this display are strictly positive. The inverse uses $t_1 = -\sqrt{B}$, $t_2 = C/t_1$, $t_3 = D/t_1$; this sign is checked against the global point.

6.4 Critical equations and printed uniqueness boxes

For either chart write $q = (q_1, q_2, q_3)$ for its three shape invariants, $E = E_n$, and $r_j^2 = R_j$. Put $w = (2, 2, 1)$ in degree six and $w = (2, 2, 2)$ in degree seven, and $L = 2 + \sum_j w_j r_j$. The six rational critical equations are

$$r_j^2 - R_j = 0 \quad (1 \le j \le 3), \qquad \sum_{i=1}^{3} w_i(\partial_{q_j} R_i - R_i \partial_{q_j} \log E) \prod_{k \ne i} r_k - 2\partial_{q_j} \log E \prod_{k=1}^{3} r_k = 0 \quad (1 \le j \le 3). \tag{90}$$

For the polynomial system, take the numerator of each rational expression after cancellation and clear its rational coefficient denominators; nonzero constant rescaling of a row is immaterial. The formula follows by differentiating $L/\sqrt{E}$ and multiplying by $2\prod r_j$. The displayed pivot and amplitude inequalities ensure that denominator clearing neither loses nor creates a solution near our point. The degree-six polynomial degrees are $(3, 4, 6, 9, 7, 6)$. For degree seven the quotient degrees are $(4, 8, 9, 15, 11, 11)$.

For the actual uniqueness checks use the degree-six invariant system and the degree-seven *unquotiented* t, r system, respectively. The latter is obtained from the same derivative formula, or explicitly by

$$r_i^2 - |P_i(t)|^2 = 0, \quad \sum_i 2\{(1 + t_j^2)\partial_{t_j}|P_i|^2 - 4t_j|P_i|^2\} \prod_{k \ne i} r_k - 8t_j \prod_k r_k = 0. \tag{91}$$

Its degrees are $(4, 8, 12, 13, 13, 13)$. Fermat's theorem applies because all real parameters in either chart give feasible normalized polynomials, the certified global point lies in an open chart, and the amplitudes are positive. It is therefore a zero of the corresponding six equations.

Here are complete rational centers for the new boxes X_n. Divide each integer in the table by 10^{12} and give every coordinate the independent radius 10^{-8}.

Degree-six coordinate	Center numerator	Degree-seven coordinate	Center numerator
B	1056172182060	t_1	-601204356501
C	364768629645	t_2	402231155999
U	422209449649	t_3	583017398799
r_1	1048247168018	r_1	1245629502750
r_2	1476040180978	r_2	1515105139677
r_3	1965146944067	r_3	1647133669279

If F_n is the six-equation polynomial map and c_n this exact center, define the preconditioner *without any additional numerical data* by $J_n = DF_n(c_n)^{-1}$ over $\mathbb{Q}$. The verifier checks this inverse exists, and proves

$$\|J_n F_n(c_n)\|_\infty < 5 \cdot 10^{-13}, \qquad \sup_{x \in X_n} \|I - J_n DF_n(x)\|_\infty < \begin{cases} 6 \cdot 10^{-6}, & n = 6, \\ 15 \cdot 10^{-6}, & n = 7. \end{cases} \tag{92}$$

Since $5 \cdot 10^{-13} + 15 \cdot 10^{-6} \cdot 10^{-8} < 10^{-8}$, $x \mapsto x - J_n F_n(x)$ maps X_n strictly into itself and contracts it. Thus X_n contains exactly one critical point. Indeed the contraction has a fixed point, J_n is nonsingular, and two zeros of F_n would be two fixed points of the same contraction. The exact global root, mapped by (82) or the peeling formulas, lies strictly inside X_n; this is an independently checked interval inclusion.

7 Univariate identification and minimality

7.1 Exact rational-univariate witnesses

The RUR witnesses have the following fully specified roles:

	$n = 6$	$n = 7$
Parameter T	S_6^2	S_7
Parameter polynomial $H(T)$	H_6, degree 186	h_7, degree 855
Coordinate order	B, C, U, r_1, r_2, r_3	B, C, D, r_1, r_2, r_3
Coordinate formula	$N_j(T)/(d_j H'(T))$	$N_j(T)/(d_j H'(T))$

Here $N_j \in \mathbb{Z}[T]$ and d_j are positive integers. These auxiliary numerator arrays belong to the machine-readable proof witness; their mathematical meaning and every condition for accepting them are given here. They do not constitute additional unspecified extremal quantities. The two witnesses are `target6_rur_exact.json` and `S7_value_RUR_EXACT.json`. In each case the verifier checks primitivity, degree, $\gcd(H, H') = 1$, the equality of the common denominator to H', and the following seven identities over $\mathbb{Z}[T]$: the six critical equations (90), and the target equation

$$\begin{aligned} 4B(2 + 2r_1 + 2r_2 + r_3)^2 - T(B+4)^2(1+U)^2(B+C^2) &= 0 \qquad (n = 6), \\ T(1+B)(B+C^2)(B+D^2) - B^2(1 + r_1 + r_2 + r_3) &= 0 \qquad (n = 7). \end{aligned} \tag{93}$$

The first represents a squared value; the second represents the value itself, with the positive signs fixed by (89).

Lemma 7.1 (Integer acceptance of a proposed RUR)**.** *Let $H \in \mathbb{Z}[T]$ be squarefree and let the proposed coordinates be $N_j/(d_j H')$. Suppose all required denominators are nonzero at a real root α of H. If every defining equation and target equation, after denominator clearing, is divisible by H in $\mathbb{Z}[T]$, the coordinates evaluated at α satisfy those equations exactly.*

Proof. Take a common denominator $D = \operatorname{lcm}(d_j)H'$ and rescale the numerators. For a polynomial equation F of total degree d the homogenized numerator is

$$\widehat{F}(T) = D(T)^d F\big(N_1(T)/D(T), \dots, N_6(T)/D(T), T\big).$$

Treating the parameter as the coordinate TD/D gives the same formula uniformly. The checked integer identity $\widehat{F} = HQ_F$, $Q_F \in \mathbb{Z}[T]$, makes $\widehat{F}(\alpha) = 0$. Divide by $D(\alpha)^d \neq 0$. □

Efficient exact verification. An expression circuit carries a pair (U, d) representing $D^d f(N/D)$. Addition first raises its operands to the same denominator degree; multiplication multiplies numerators and adds degrees; integer powers act accordingly. Common subexpressions are evaluated once and released after their final use. The verifier accepts divisibility only after constructing a quotient Q_F and checking the *integer* product $HQ_F = \widehat{F}$. A quotient proposed by arithmetic modulo a large prime power is merely a speed improvement; incorrect proposals are rejected by the exact multiplication. No modulus-size heuristic replaces that final identity. For S_7 the largest homogeneous degree is 15, giving a numerator degree 12810. The seven checks are processed sequentially to control memory.

7.2 Unique univariate roots and selection of the real branch

For a real polynomial p and $I = (c-r, c+r)$, opposite endpoint signs and $0 \notin p'([c-r, c+r])$ prove existence and uniqueness of a simple root by the intermediate value theorem and strict monotonicity. A convenient sufficient rational test, for $p(c+y) = \sum b_j y^j$ and $r > 0$, is

$$p(c-r)p(c+r) < 0, \qquad |b_1| > \sum_{j=2}^{\deg p} j|b_j|r^{j-1} \tag{94}$$

Under these inequalities, $(c-r, c+r)$ contains exactly one simple root: the second inequality keeps the derivative of one sign, and the first gives existence. All endpoint signs and derivative bounds are checked either as rational inequalities or with outward Arb balls. For h_6 this test is checked entirely in rational arithmetic on I_6. For h_7 the opposite endpoint signs and exclusion of zero from the derivative interval are checked directly on I_7 after a rigorously rounded Taylor shift, at precision exceeding the coefficient height plus $4(856) + 512$ bits. The derivative signs on the stated intervals are also recorded by the verifier. This establishes uniqueness in the *printed rational intervals*, rather than merely in an unreported smaller neighborhood.

For branch evaluation we refine the unique parameter root by interval Newton on the Taylor-shifted polynomial. Degree six starts with (a^2, b^2), where $(a, b) = I_6$, and checks the same unique-root criterion for H_6 on that explicit squared interval. Degree seven starts with I_7. Thus no unreported starting interval or numerical approximation is needed. Refinement is used to evaluate the RUR reliably. The verifier proves that $H'(T)$ and all chart pivots are separated from zero and all r_j are positive.

Proposition 7.2 (Identification with the unrestricted global optimum)**.** *The RUR branch selected in the preceding paragraph has target $T = S_6^2$ for degree six and $T = S_7$ for degree seven.*

Proof. The phase certificates give the unrestricted global extrema $C_n = S_n$. Their reciprocal representatives map into the explicit boxes X_n of Section 6.4, and are stationary because the normalized rational charts give feasible polynomials for every nearby real parameter. For degree six the RUR gives (B, C, U, r_1, r_2, r_3) directly. For degree seven use $t_1 = -\sqrt{B}$, $t_2 = C/t_1$, $t_3 = D/t_1$; the nonzero Jacobian in (88) and the chain rule convert its exact quotient critical equations to (91).

Strict interval evaluation places this second critical point inside the same X_n. Its uniqueness proved in (92) identifies the two points exactly. Finally (93), the energy formulas and positivity of the amplitudes give $T = C_6^2$ or $T = C_7$. These steps use the global cover, critical-point uniqueness, and exact RUR identities separately; matching decimal expansions alone would not identify the algebraic number. □

7.3 Irreducibility and the exact degrees

For h_5, reduction modulo 7 and multiplication by its leading coefficient inverse gives

$$\bar{h}_5 = X^4 + X^3 + 3X^2 + 2X + 4 \quad \text{in } \mathbb{F}_7[X].$$

The exact Frobenius checks $X^{7^4} = X \pmod{\bar{h}_5}$ and $\gcd(X^{7^2} - X, \bar{h}_5) = 1$ prove irreducibility. For completeness, the latter remainder is $3X^3 + 3X^2 + X + 6$; its gcd with $\bar{h}_5$ is one. Gauss's lemma proves the degree-four assertion in Theorem 1.1.

We recall the degree-intersection argument used for the larger polynomials. For a primitive degree-d integer polynomial h and a prime preserving its degree with squarefree reduction, every rational factor degree is a subset sum of the degrees of the irreducible factors modulo that prime. Indeed Gauss's lemma gives integer factors, and preservation of the product's leading coefficient preserves both factor degrees modulo the prime. Intersecting these possible-degree sets over several primes can therefore prove irreducibility without finding a single irreducible reduction. Each modular factor is independently checked: for a monic factor g of degree e, check

$$X^{p^e} = X \pmod{g}, \qquad \gcd(X^{p^{e/q}} - X, g) = 1 \quad (q \text{ prime},\ q \mid e). \tag{95}$$

The criterion follows because the roots of $X^{p^k} - X$ are exactly the elements of $\mathbb{F}_{p^k}$, and an irreducible polynomial of degree f divides it exactly when $f \mid k$. The proper divisors of e are all divisors of at least one e/q. The verifier also checks primehood of every displayed prime, squarefreeness, and the complete factor product.

For H_6 the verified modular degree patterns are

Prime	Irreducible factor degrees of H_6
1000033	$1, 5, 6, 7, 67, 100$
1000037	$3, 21, 21, 46, 95$
1000039	$1, 1, 3, 3, 6, 12, 160$

Their subset-sum intersection is exactly $\{0, 186\}$. Hence H_6 is irreducible. Let α be its selected positive root. The rational field norm is $N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha) = H_6(0)/\operatorname{lc}(H_6)$. Its 5-adic valuation is -75, as checked by integer division of those two coefficients. It is therefore not a rational square. If α were a square in $\mathbb{Q}(\alpha)$, its norm would be a rational square. Consequently

$$[\mathbb{Q}(\sqrt{\alpha}) : \mathbb{Q}(\alpha)] = 2, \qquad [\mathbb{Q}(\sqrt{\alpha}) : \mathbb{Q}] = 372.$$

Thus $H_6(X^2)$, of degree 372, is irreducible. By Proposition 7.2, $\sqrt{\alpha} = S_6 > 0$, proving Theorem 1.2 with the root test on I_6.

For h_7 the verified patterns are

Prime	Irreducible factor degrees of h_7
1000033	$1, 1, 21, 38, 150, 278, 366$
1000037	$1, 1, 8, 25, 51, 769$
1000039	$1, 5, 112, 228, 509$
1000081	$5, 116, 145, 589$

Their subset-sum intersection is $\{0, 855\}$. Thus h_7 is irreducible; its largest coefficient has 2641 bits. Proposition 7.2 and the root test on I_7 prove Theorem 1.3. These arguments also complete the minimal-polynomial part of Theorem 1.1; its feasibility and global upper bound were proved in Section 4.

8 Structural consequences and a reusable proof procedure

8.1 The square does not lower the degree for S_7

Corollary 8.1. *One has $\mathbb{Q}(S_7) = \mathbb{Q}(S_7^2)$. Write $h_7(X) = A(X^2) + XB(X^2)$. Then*

$$g(Y) = \operatorname{prim}\{YB(Y)^2 - A(Y)^2\}$$

is the degree-855 integer minimal polynomial of S_7^2, and $S_7 = -A(S_7^2)/B(S_7^2)$.

Proof. The degree of $\mathbb{Q}(S_7)$ over $\mathbb{Q}(S_7^2)$ is at most two and divides the odd integer 855, so it is one. The odd degree of h_7 makes B nonzero. Since $[\mathbb{Q}(S_7^2) : \mathbb{Q}] = 855$ and $\deg B < 855$, one has $B(S_7^2) \neq 0$. The displayed relation gives the rational recovery. The polynomial annihilator has degree 855, so field degree proves its minimality. The verifier additionally checks $h_7(X)h_7(-X) = -g_0(X^2)$ for the unnormalized expression $g_0(Y) = YB(Y)^2 - A(Y)^2$, including its coefficient normalization. □

8.2 The normal closure of the degree-seven value

Corollary 8.2. *The splitting-field Galois group of h_7 over $\mathbb{Q}$ is $\mathfrak{S}_{855}$. Thus S_7 is not expressible by radicals over $\mathbb{Q}$, the root field $\mathbb{Q}(S_7)$ has no proper intermediate fields over $\mathbb{Q}$, and its normal closure has degree* 855!.

Proof. Irreducibility gives a transitive Galois group on 855 roots. The squarefree factorization at 1000037 gives a Frobenius permutation of cycle type $(1, 1, 8, 25, 51, 769)$. Its 10200th power is a single 769-cycle fixing 86 points, since 769 is prime and coprime to $10200 = \operatorname{lcm}(8, 25, 51)$. A nontrivial block system has both its number of blocks and their sizes at most $855/2 < 769$. An element of prime order 769 would then act trivially on the blocks and within each block, a contradiction. The group is primitive. Jordan's prime-cycle theorem [5, Theorem 1.1] gives $\mathfrak{A}_{855} \subseteq G$. The original permutation is odd, so $G = \mathfrak{S}_{855}$. The symmetric group is not solvable, giving the radicals assertion. A root stabilizer is $\mathfrak{S}_{854}$, a maximal subgroup of $\mathfrak{S}_{855}$; Galois correspondence gives the intermediate-field assertion. The splitting field has degree $|G| = 855!$. □

8.3 What is reusable, and what requires new mathematics

Theorem 8.3 (Local identification without complete elimination)**.** *Suppose a separate global argument gives an extremizer P_* and a local chart whose critical coordinate x_* lies in a rational convex box X. Assume its critical polynomial map F is injective on X. Let H be a primitive irreducible integer polynomial with positive leading coefficient and exactly one root α in a rational interval I. Suppose a proposed rational coordinate map $R(T)$ satisfies $F(R(T)) = 0$ and $T = V(R(T))$, where V is the desired target function, by the integer acceptance test of Lemma 7.1, and interval evaluation proves $R(\alpha) \in X$ with all denominators nonzero. Then the target value at P_* equals α. In particular H is its integer minimal polynomial. No description of the other components of the complex critical variety is needed.*

Proof. The integer identities give $F(R(\alpha)) = 0$, while stationarity gives $F(x_*) = 0$. Injectivity identifies the two points in X. The exact target equation identifies their target values, and irreducibility gives minimality. If the target is a squared value, the nonsquare-norm test used for S_6 supplies the additional degree-two extension. □

The reusable verifier consists of four interfaces: an unrestricted global certificate, an explicit local rational chart with checked nonzero pivots, an exact RUR satisfying all integer identities, and a real uniqueness box containing both the global point and the RUR branch. A target polynomial and an irreducibility witness complete the output. Taking a sign quotient before elimination reduces the system while preserving the local branch; homogeneous integer circuits make verification substantially less expensive than rediscovering a lexicographic elimination basis.

Lemma 8.4 (Target recovery from traces)**.** *Let A be a finite reduced algebra over a perfect field k of dimension d, and suppose $T \in A$ separates its d geometric points. For $m_j = \mathrm{Tr}(M_T^j)$, the least linear recurrence of $(m_j)_{j\geq 0}$ has order d and characteristic polynomial equal to the minimal polynomial of T. It is determined by $m_0, \ldots, m_{2d-1}$.*

Proof. Over an algebraic closure, A is the product of d copies of the field and M_T has distinct eigenvalues $\alpha_1, \ldots, \alpha_d$. Thus $m_j = \sum_i \alpha_i^j$. The d by d Hankel matrix $(m_{i+j})_{0\leq i,j<d}$ is $V^T V$, where V is the Vandermonde matrix on the distinct α_i; its determinant is nonzero. The coefficients of the monic recurrence of order d are therefore uniquely obtained from the d equations with indices $0, \ldots, d-1$. The product $\prod_i (X - \alpha_i)$ satisfies them. The nonzero Hankel determinant rules out a smaller recurrence, proving the assertion. □

An effective discovery route uses arithmetic modulo successive good primes. On a finite reduced component, traces of powers of multiplication by a separating target determine a linear recurrence for its powers; its minimal polynomial is recovered from that recurrence. Chinese remaindering and rational reconstruction propose characteristic-zero coefficients, followed by Newton lifting of the coordinate RUR. These are discovery procedures: no claimed stabilization, generic component degree, or probabilistic reconstruction is evidence until Lemma 7.1, the real bridge, and irreducibility have been checked.

Proposition 8.5 (Conditional finite verification search)**.** *Fix n and a feasible algebraic candidate C. Suppose a prescribed compact set $\mathcal{M}$ of candidate phase points is covered by finitely many effective locally sharp neighborhoods, and $L_n(b) < C$ outside $\mathcal{M}$. A fair search through rational positive Toeplitz majorants and finite rational phase-box covers eventually certifies $S_n = C$.*

Proof. At $b \notin \mathcal{M}$, the exact complex dual gives a majorant with diagonal below C^2. Add a sufficiently small positive scalar matrix and approximate its entries rationally: strict positivity and the strict diagonal bound persist. They continue to hold on a neighborhood of b. Together with the local sharp neighborhoods these open sets cover the compact phase torus. A sufficiently fine finite box cover is subordinate to a finite subcover; all strict inequalities terminate at finite outward precision. Fair enumeration eventually reaches such a finite witness. □

Strict separation outside $\mathcal{M}$ is a hypothesis of this conditional statement; it has not been established for an unknown degree. The abstract criterion does not require full contact or symmetric extremizers. The general algebraicity and strict-growth theorems hold without these hypotheses, whereas efficient exact closure in a new degree requires further structural information or a different global certificate.

A Complete rational stationary boxes

For $n = 6, 7$, the six ordered groups are those in (68). Each displayed integer is divided by 10^{15}; each coordinate is independently restricted to the closed interval of radius 10^{-12} about that rational center. Thus this appendix, together with the rational equations (67), defines P_n^* and the intermediate values C_n *exactly*; no decimal values are used in their definition. Each verifier checks the existence and uniqueness of the root, including its positivity and contact nondegeneracy, with outward integer intervals.

Degree six: 37 coordinates

```
x:
  262107422046759, 274753362877041, -56834182227533 ,
  21931482111926, 200254489084666, 192750050669383 ,
  -97448045755226
y:
  382683747964617, 514612479312783, -331021320804505 ,
  195797927412915, -243319088997319
u:
  975367953712551, 197013293293456, -814439100551040 ,
  -943144476618461, -197013293293456, 741528708673153
v:
  220584121982048, 980400817148612, 580249042647735 ,
  -332383056463608, -980400817148612, -670921138594936
lambda:
  433120892734887, 310501280533748, 433120892734887 ,
  325904972998054, 534010330647529, 325904972998054
rho:
  262107422046759, 274753362877041, 386881086673566 ,
  515079599452430, 386881086673566, 274753362877041 ,
  262107422046759
```

Degree seven: 43 coordinates

```
x:
  235920876276069, 293870003804017, 236641874895934 ,
  388564195563446, -96883585759302, -19530465759307 ,
  -284748638818354, 226824433433431
y:
  -267894573672937, -4749259739664, 376322016831812 ,
  -356910969941934, 72648412418080, 64879397812180
u:
  780369667645909, -358685493593095, -914405693017847 ,
  -918268518904145, -261135887151378, 593409122120395 ,
  993774896189023
v:
  625318464318953, 933458470788003, 404798997746474 ,
  -395958239199021, -965302050366445, -804900996262461 ,
```

```
  -111406713013609
lambda:
  364962313977169, 364962313977169, 354670796377332 ,
  390267013871127, 331857813315176, 390267013871127 ,
  354670796377332
rho:
  235920876276069, 293870003804017, 357444932203085 ,
  388593218600046, 388593218600046, 357444932203085 ,
  293870003804017, 235920876276069
```

B Rational matrices for the moving-atom certificates

All entries of A, B, L, W below are the displayed integers divided by 10^8. The contact rows are in the order of Appendix A; the internal phase columns are $1, \dots, n-1$. Thus these arrays, the stationary boxes, and formulas (72)–(76) specify all small locally sharp matrices without relying on a numerical optimizer. The normalized center phase is computed from the candidate coefficients using $b_k = \bar{a}_k/|a_k|$ and the endpoint gauge.

Degree 6

$$A = 10^{-8}\begin{pmatrix} -14890744 & -10643183 & 10527552 & 430039 & 7265783 \\ 8032646 & 7028881 & -1731438 & 2596755 & -7365074 \\ -7430126 & -10677968 & -4325848 & -9269467 & 15001693 \\ 13193678 & 36775846 & -7224118 & 5874034 & 15161723 \\ -7284904 & -27793980 & 989159 & 7050311 & 17510364 \\ 828684 & 629274 & 3790850 & -10548300 & -41409359 \end{pmatrix}.$$

$$B = 10^{-8}\begin{pmatrix} 71907201 & 94764809 & -18713451 & 16091082 & 44903955 \\ 0 & 72850058 & 2977426 & 12459542 & 27992069 \\ 0 & 0 & 35688982 & 9762342 & -6075938 \\ 0 & 0 & 0 & 46124378 & 64205450 \\ 0 & 0 & 0 & 0 & 129478077 \end{pmatrix}.$$

$$L = 10^{-8}\begin{pmatrix} -22294792 & 18920044 & 46526017 & -25458834 & 11736207 \\ 40067266 & -14185195 & -3574022 & -42663677 & 10786177 \\ 37456900 & -11829466 & 26079253 & 39216871 & -4374547 \\ -23362184 & 6269030 & -16019440 & 17186319 & -5659321 \\ -6181973 & -5697849 & -34293857 & -9172402 & -5770265 \\ -25685217 & 6523436 & -18717951 & 20891723 & -6718251 \end{pmatrix}.$$

$$W = 10^{-8}\begin{pmatrix} 117169743 & 805182 & 2077663 & -1643164 & -706629 & -19934344 \\ 5025575 & -31126564 & 23613388 & 10488322 & 14397646 & 4817360 \\ -5146156 & -12064513 & -22326302 & -27169639 & 10844357 & 4426359 \\ 0 & 0 & 6232805 & 0 & -30931398 & 8499186 \\ 5146156 & 12064513 & -22326302 & 27169639 & 10844357 & 4426359 \\ -5025575 & 31126564 & 23613388 & -10488322 & 14397646 & 4817360 \\ -117169743 & -805182 & 2077663 & 1643164 & -706629 & -19934344 \end{pmatrix}.$$

Degree 7

$$A = 10^{-8}\begin{pmatrix} -9444419 & -1659413 & 5149452 & -15278593 & 6765714 & 36534907 \\ -12797789 & -22954099 & 6848662 & 1688835 & 19457058 & 25205002 \\ 17094407 & 18382667 & 14276317 & -3035961 & -25548582 & -13109865 \\ -422333 & 5224977 & -11510711 & -1743947 & -16980425 & 14091864 \\ 11159133 & 4887015 & -6533311 & 4263583 & 2286339 & -5553008 \\ -13202917 & -1745721 & 392253 & 13677309 & 15553000 & -6060643 \\ 1900218 & -8673863 & -1790713 & -6552626 & 3156487 & -39190360 \end{pmatrix}.$$

$$B = 10^{-8}\begin{pmatrix} 65381905 & 63215613 & 24838007 & -2331180 & -129568710 & -81806523 \\ 0 & 61218102 & 35016321 & 2957669 & -84069385 & -124386129 \\ 0 & 0 & 71134781 & 19563617 & -27915195 & -75725631 \\ 0 & 0 & 0 & 61050283 & 64229603 & 63811655 \\ 0 & 0 & 0 & 0 & 82452000 & 144523872 \\ 0 & 0 & 0 & 0 & 0 & 179420222 \end{pmatrix}.$$

$$L = 10^{-8}\begin{pmatrix} 6351281 & 1605841 & 37379964 & -21944470 & 9116688 & 10642255 \\ -6726643 & 32104528 & -11781454 & 26406732 & -4109303 & -4019270 \\ 19825778 & -27910010 & 10620929 & 12276733 & -14079378 & -3962551 \\ -40288952 & -7910120 & -17863047 & -18337565 & -25834166 & -11535375 \\ -13194733 & 32649882 & 19642480 & -4684989 & 26973448 & -8614817 \\ 32276071 & -4139460 & -28141687 & -30147003 & 22379803 & 3086215 \\ 1757198 & -26400661 & -9857185 & 36430562 & -14447092 & 14403543 \end{pmatrix}.$$

$$W = 10^{-8}\begin{pmatrix} 60496914 & 5336249 & -1302144 & 4740863 & 1572240 & 249130 & 19894126 \\ 5978239 & -25052266 & -22030264 & -3873332 & 18093299 & -8427224 & -4596516 \\ -1321300 & -19226778 & 16013454 & 19565782 & -11042330 & -14970233 & -3845304 \\ -13634960 & 14555414 & 2720863 & 17440436 & 15978924 & 19992030 & -5064878 \\ 13634960 & -14555414 & 2720863 & -17440436 & -15978924 & 19992030 & -5064878 \\ 1321300 & 19226778 & 16013454 & -19565782 & 11042330 & -14970233 & -3845304 \\ -5978239 & 25052266 & -22030264 & 3873332 & -18093299 & -8427224 & -4596516 \\ -60496914 & -5336249 & -1302144 & -4740863 & -1572240 & 249130 & 19894126 \end{pmatrix}.$$

C The independent finite verifier

The independent degree-five global program is `Sidon5_Exact_Closure_Certificate_Revised.py`. It is distributed beside this LaTeX source and runs with the command

```
python3 Sidon5_Exact_Closure_Certificate_Revised.py
```

without a solver, third-party libraries, network access, or prior computed files. Assertions are part of the verification, so Python must not be run with `-O`. All real intervals are represented by two signed arbitrary-precision integers $[L, U]$ corresponding to $[L/10^{50}, U/10^{50}]$. Addition, subtraction, multiplication, positive division, square, and nonnegative square root round their endpoints outward; the last operation uses the integer square root. Complex intervals are pairs of real intervals.

The analytic inputs passed to the verifier are precisely these:

1. Arithmetic in $\mathbb{Q}[w]/(16w^4+10w^3-8w+1)$ proves (2), (20), $P_*(1) = 1$, the sign pattern, $T_*p = \Lambda s$, and $T_*q = 0$. Rational root isolation bounds the associated algebraic coefficients.

2. Real LDL^T tests verify $N^TN \succ 0$, the gap (30), invertibility of $\mathcal{B}$ and FF^T, and the positive quadratic matrix (38). The code constructs the minimizer and the polarization tensors from (33)–(47), then verifies (38), (44), (49), and (56), using rational intervals.

3. The program reconstructs π from $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ with alternating Taylor bounds (50 terms for each arctangent), and roots of unity by repeated half-angle square roots and complex multiplication. The identity for π follows from the tangent addition formula and the fact that $0 < 4\arctan(1/5) - \arctan(1/239) < \pi/2$ has tangent one. Thus every trigonometric enclosure is rigorous.

4. The global payload is decoded from base85 and zlib and checked against the displayed SHA-256 digest. A depth-first traversal of each of the 262660 initial orbit representatives reads exactly one tag per node and, for Toeplitz leaves, ten signed 64-bit integers representing five complex moments. At the end the byte stream must be exhausted and all expected counts must match. Every Fourier, local and Toeplitz leaf is independently rechecked against the corresponding strict inequality. Toeplitz positive definiteness is proved by interval complex LDL^*, requiring the lower endpoint of each pivot to be positive.

For avoidance of ambiguity, the recursion implemented in the file has the following mathematical pseudocode. All tests are *strict* comparisons of enclosing intervals; failure at any assertion invalidates the certificate.

```
VERIFY-BOX(level, t):
  read next tag; reject missing or invalid tag
  if tag = FOURIER:
    compute interval enclosure of DFT mass at center
    assert upper(mass) < lower(L0 cos(h[level]))
  else if tag = LOCAL:
    assert a ten-center ball contains the complete box
  else if tag = TOEPLITZ:
    read ten integers; construct the six-by-six Hermitian
    Toeplitz matrix with diagonal (L0 cos(h[level])) squared
    assert every interval LDL-star pivot of T-R(center) is positive
  else if tag = SPLIT:
    for all 16 sign vectors with entries -1 or +1:
      VERIFY-BOX(level+1, exact-child(level,t,sign))
  else:  reject
FOR each lexicographic initial orbit representative t:
  VERIFY-BOX(0,t)
assert all bytes consumed, node counts and layer counts exact.
```

For a Hermitian matrix A, the interval LDL^* recurrence uses

$$d_j = A_{jj} - \sum_{k<j} |L_{jk}|^2 d_k, \qquad L_{ij} = \frac{A_{ij} - \sum_{k<j} L_{ik}\overline{L_{jk}} d_k}{d_j},$$

where each positive divisor has a strictly positive certified lower bound. By induction, the true pivots are contained in the computed intervals. Six positive pivots certify $A \succ 0$, with no floating-point eigenvalue approximation. The minimum reported pivot is an LDL^* pivot lower bound, not a uniform lower bound for all minimum eigenvalues.

The payload is compressed solely for file size: decoding is deterministic, and the verifier does not accept its recorded labels or moments without recomputing all associated inequalities. In particular, the digest authenticates one dataset but is not itself a proof of the mathematical bounds. The proof obligation is jointly discharged by the arguments of Sections 3 and 4 and the full execution of the independent verifier.

D Complete certificate interface and pseudocode

The archive `Exact_Sidon_5_to_7_Proof_Certificates.zip` contains exactly the proof data needed for the three closures and the stated structural corollaries. From its extracted directory run

```
python3 -m pip install -r requirements.txt
python3 verify_all.py
```

Use Python 3.10 or later, and do not use `-O`: assertions are proof predicates. The pinned algebraic dependencies are `python-flint==0.9.0` and `sympy==1.14.0`. The verifier makes no network calls. Dependency installation is a setup step, not a mathematical premise. Separate options `--part global` and `--part algebraic` rerun just those layers; the algebraic-only mode does not assert that it has rerun the complete global cover.

Proof component	Executable or data interface
S_5 feasibility, sharp local matrices and full phase cover	`Sidon5_Exact_Closure_Certificate_Revised.py`
S_6, S_7 unrestricted stationary roots, local atoms, covers and reversal	`Sidon6_7_Exact_Closure_Certificate.py`
New six-variable boxes and global-to-chart bridge	`critical_bridge.py`, `critical_boxes.json`
S_6 RUR, degree 372, interval and norm obstruction	`verify_s6_algebraic.py`, `target6_rur_exact.json`
S_7 RUR, degree 855, interval and modular sieve	`certify_s7_value_rur.py`, `S7_value_RUR_EXACT.json`
Characteristic-zero divisibility identities	`integer_circuit_verify.py`
Primitivity, exact root tests and finite-field factor checks	`exact_values.py`, `irreducibility.py`
Long polynomial displays, checked against both RURs	`S6_S7_Minimal_Polynomials.txt`

D.1 The complete global step

The following pseudocode specifies the recursion for degrees six and seven; Appendix C gives the specialized degree-five format. All comparisons refer to lower and upper endpoints of verified enclosures. Coverage includes the boundaries of adjacent cells.

```
VERIFY_GLOBAL(n):
    certify the stationary root and its Cartesian rational box;
    prove positive weights, positive amplitudes, distinct contacts;
    check the positive constant in the complementary identity;
    reconstruct A,B,L,W and all local Taylor--Schur bounds;
    check the normalized reversal root returns to its uniqueness box;
    for every canonical initial phase-orbit representative:
        VERIFY_NODE(initial_box);
    assert complete data exhaustion and exact traversal statistics.

VERIFY_NODE(box):
    read and validate the next tag;
    if tag is DFT or INTERPOLATION:
        rebuild the roots of unity and the interpolation weights;
        assert upper(center_bound) < lower(aligned_threshold);
    else if tag is TOEPLITZ:
        rebuild the rational Toeplitz matrix;
        assert every enclosing LDL-star pivot has positive lower end;
        assert diagonal <= aligned_threshold squared;
    else if tag is LOCAL:
        assert the entire box lies in a proved local ellipsoid;
    else if tag is BRANCH:
        construct all 2**(n-1) exact children;
        VERIFY_NODE(child) for every child, in the specified order;
    else: reject.
```

The global source root need not have full contact merely because it is stationary; this property is verified for the designated candidate only. Arbitrary competitors are excluded by the full phase

cover and are not restricted to a full-contact or palindromic stratum.

D.2 The algebraic step and the final identification

```
VERIFY_ALGEBRAIC(n):
    reconstruct the exact rational critical equations from the chart;
    read H and the six rational coordinate numerators N_j/(d_j H');
    assert characteristic zero, primitive H, degree and denominator;
    prove that the printed value interval has exactly one root;
    verify every modular factor by Frobenius powers and gcd tests;
    intersect all possible rational factor-degree sets;
    if n == 6: prove the nonsquare field norm and degree doubling;
    for each of the six critical equations and the target equation:
        form the denominator-cleared integer circuit numerator U;
        propose Q; assert the exact integer identity H*Q == U;
    check the symbolic chart identities and inverse pivot conditions;
    verify the printed critical box by exact-center preconditioning;
    map the unrestricted global root into that box;
    refine the unique real parameter root by interval Newton;
    evaluate every RUR coordinate with outward rounding;
    assert nonzero denominators and the positive amplitude branch;
    map that point into the same critical box;
    deduce equality of points by uniqueness and apply the target equation;
    compare every coefficient and interval in the polynomial TXT;
    accept the algebraic layer.

VERIFY_ALL():
    VERIFY_GLOBAL(5); VERIFY_GLOBAL(6); VERIFY_GLOBAL(7);
    check h_5 is primitive and irreducible modulo 7;
    VERIFY_ALGEBRAIC(6); VERIFY_ALGEBRAIC(7);
    verify the odd-degree square and Galois-group corollaries;
    accept all three complete exact closures.
```

The actual global program for degrees six and seven shares its root and matrix infrastructure and runs both covers in one invocation. The integer circuits are verified one at a time; parallel evaluation of the largest degree-seven numerators would unnecessarily multiply memory use.

The standalone source files, full global payloads, RUR arrays, modular factors and reproducibility logs are distributed together. Compressed payloads and SHA-256 manifests detect accidental changes; mathematical acceptance is always by the reconstructed predicates above. The trusted arithmetic consists of arbitrary-precision integers and rationals, FLINT polynomial arithmetic, and outward-rounded Arb enclosures. This is a computer-assisted proof with explicit finite obligations; we do not claim a proof-assistant formalization of the three closures.

Theorem 1.4 requires no execution of this archive: its complete proof is Lemma 2.1, Lemma 2.2 and the subsequent extremizer argument.

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